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Methods for purifying proteins having a tubulin carboxypeptidase activity and peptidic based inhibitors thereof

Abstract

The invention relates to a method for purifying proteins having a tubulin carboxypeptidase activity from a biological extract, comprising polymerization/depolymerization cycle performed on a biological extract in presence of microtubules. The invention further relates to a peptidic based inhibitor for use in the treatment of a disorder involving altered microtubule detyrosination in an animal, wherein the peptidic based inhibitor comprises a peptidic moiety constituted of 1 to 20 amino acids, said peptidic moiety having an amino acid selected from Y or F at the C-terminal position, and wherein the peptidic based inhibitor inhibits at least partially a tubulin carboxypeptidase activity.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

- (1) This application is the U.S. national stage application of International Patent Application No. PCT/EP2018/069496, filed Jul. 18, 2018.
- (2) The Sequence Listing for this application is labeled "Seq-List-replace.txt" which was created on Feb. 11, 2020 and is 62 KB. The entire content of the sequence listing is incorporated herein by reference in its entirety.

FIELD OF THE INVENTION

- (3) The present invention relates to a method for purifying a protein having a tubulin carboxypeptidase activity in a biological extract. The invention further relates to a method for identifying a peptidic based inhibitor suitable for inhibiting a tubulin carboxypeptidase activity. The present invention also relates to the use of such peptidic based inhibitors for use in the treatment of a disorder involving defects in microtubule detyrosination in an animal, preferably a mammal.

BACKGROUND OF THE INVENTION

- (4) Microtubules (MTs) are the major types of filaments that constitute the eukaryotic cytoskeleton. They are formed by the polymerization of a dimer of two globular proteins, α - and β -tubulin heterodimers. They are involved in many different functions including intracellular transport (cargo transport), cell motility, cell division, cell morphogenesis and convey mechanical signals to intracellular effectors (mechanotransduction). The intrinsic dynamic instability of the cytoskeletal microtubular system is essential for neuronal remodelling, plasticity and adaptation. Each particular MT function requires the recruitment of a specific set of MT-associated proteins (MAPs) and molecular motors. Many MAPs and motors interact with the C-terminal tails of tubulins, which protrude from the MT surface (Ciferri et al, 2008; Mizuno et al, 2004; Roll-Mecak & Vale, 2008; Skiniotis et al, 2004). Thus, one way to adapt MTs to different functions is to change the properties of the tubulin C-terminal tails through posttranslational modifications.
- (5) Among said post-translational modifications of the tubulin C-terminal tails, two polymodifications, namely polyglutamylation and polyglycylation, occur on both α - and β -tubulin (Edde et al, 1992; Redeker et al, 1994). Polyglutamylation and polyglycylation consist of the addition of side chains composed of either glutamate or glycine residues to the primary sequence glutamates present at the C-terminus of both tubulins. Enzymes that catalyze these modifications have recently been identified (Janke et al, 2005; Rogowski et al, 2009; van Dijk et al, 2007) as well as the enzymes that remove polyglutamylation (Rogowski et al, 2010). Apart from polymodifications, also detyrosination occurs on the C-terminus but it is specific to α -tubulin (Arce et al, 1975). Detyrosination consists of the removal of the very C-terminal

tyrosine from α -tubulin and its generation of so-called $\Delta 1$ -tubulin (FIG. 1).

(6) Up to now, the enzymes responsible for detyrosination that possess tubulin carboxypeptidase (TCP) activity are unknown. Of note, native TCP activity contained in protein extracts obtained from different tissues and/or organisms likely differ in their set of enzymes responsible for detyrosination.

(7) The identification of specific inhibitors of enzymes with TCP activity is of particular interest for treating disorders involving microtubule detyrosination, such as neurodegenerative diseases, neuronal regeneration disorders, cancers, muscular dystrophies, heart diseases, vascular disorders, retinal degeneration, infertility or ciliopathies.

(8) Accordingly, there is thus a need for a method allowing identification of enzymes with TCP activity and a method to allow design and identification of molecules that act on native TCP enzymatic activity.

SUMMARY OF THE INVENTION

(9) Interestingly, the present invention proposes methods for identifying both, the enzymes that possess TCP activity and peptidic based inhibitors regardless of the original tissue and/or organism. To this end, the present invention proposes to purify TCP activity from a biological extract and to use such purified biological extract, which exhibits native TCP activity to test and identify peptidic based inhibitors. More particularly, the inventors surprisingly discovered that it is possible to exploit the natural substrate of tubulin carboxypeptidases, i.e. the very C-terminal amino acid(s) of α -tubulins, as moieties or backbone for modifications, to inhibit TCP activity. The peptides that share sequence homology with the C-terminal amino acid sequence of α -tubulins could mimic the natural substrate of enzymes that possess TCP activity and consequently inhibit its activity. More particularly, the inventors have inventively used peptides composed of variable lengths (1 to 20 amino acids) of the very C-terminal sequence of α -tubulin that protrudes out of the hollow tube comprised by the MT structures, to inhibit native TCP activity contained in biological extracts. Highly selective, cell permeable, reversible or irreversible (suicide ligands) modified peptides with undetectable cytotoxicity that specifically modulate TCP activity were generated. The present invention further proposes a group of chemically modified peptides that pharmacologically act on microtubule dynamics by specifically modulating TCP activity.

(10) Since the enzymes responsible for detyrosination that possess tubulin carboxypeptidase (TCP) activity may be valuable pharmacological targets, the present invention now proposes a method for identifying enzymes responsible for detyrosination that possess tubulin carboxypeptidase (TCP) activity. To do so, the inventors originally set up a biochemical approach to isolate TCP activity contained in a biological extract followed by a specific enrichment of all proteins that could bind to microtubules. As such, the set of microtubule associated proteins (MAPs) was identified before and after the biochemical purification steps. It is thus an object of the present invention to provide a method for purifying proteins having a tubulin carboxypeptidase activity from a biological extract, comprising: (a) centrifuging the biological extract at a temperature comprised between 0 and 10° C., preferably between 2 and 5° C., more preferably at 2° C.; (b) recovering the supernatant from step (a) and proceeding to a first microtubule polymerization cycle by adding GTP and incubating the mixture at a temperature between 35 and 40° C., preferably at 37° C., $\pm$ 2° C., then centrifuging; (c) recovering the pellets of step (b), resuspending in ice-cold buffer, incubating at 4° C. $\pm$ 1° C., and proceeding to a second microtubule polymerization cycle by adding GTP and incubating the mixture at 37° C., $\pm$ 2° C., then centrifuging; (d) recovering the pellets of step (c) resuspending in ice-cold buffer, incubating at 4° C. $\pm$ 1° C., and proceeding to a third microtubule polymerization cycle by adding GTP and incubating the mixture at 37° C., $\pm$ 2° C., then centrifuging; (e) resuspending the pellets of step (d) and submitting the mixture to an ion exchange chromatography and recovering the flow through; (f) precipitating the proteins of the flow through with a 60% saturated ammonium sulphate solution; (g) submitting the precipitated fraction of step (f) to an hydrophobic chromatography and eluting by gradually decreasing ammonium sulphate concentration up to zero to recover the fraction of proteins with a tubulin carboxypeptidase activity.

(11) It is a further object of the invention to provide a method for selecting a peptidic based inhibitor able to inhibit a tubulin carboxypeptidase activity among peptidic based inhibitor candidates that comprise a peptidic moiety constituted of 1 to 20 amino acids, said modified peptide having at the C-terminal position an amino acid selected from Y or F, wherein the method comprises (a) contacting the peptidic based inhibitor candidate with a mixture containing both a fraction of native or recombinant proteins with a tubulin carboxypeptidase activity and microtubules, which preferably comprise synthetic microtubules

- and/or α -tubulins, with labeled C-terminal Y; and (b) measuring the level of isolated Y and/or detyrosinated microtubules.
- (12) Advantageously, the fraction of proteins with a tubulin carboxypeptidase activity is obtained with the method for purifying proteins as exposed above.
- (13) In an embodiment, the microtubules comprise synthetic microtubules and/or α -tubulins, with labeled C-terminal Y.
- (14) In an embodiment, the level of isolated Y in the reaction sample is compared to the level of isolated Y in a control sample comprising solely a fraction of proteins with a tubulin carboxypeptidase activity and microtubules.
- (15) In a particular embodiment, the peptidic moiety of the peptidic based inhibitor candidate is constituted of between 1 and 20 amino acids of the most C-terminal amino acids of an alpha-tubulin.
- (16) In a particular embodiment, the peptidic moiety of the peptidic based inhibitor candidate is constituted of between 1 and 16 of the most C-terminal amino acids of the amino acid sequence Nter-X1-X2-X3-X4-X5-X6-X7-X8-X9-X10-X11-X12-X13-X14-X15-X16-Cter, wherein X1, X2, X5, X7, X9 and X13 are hydrophobic amino acids, preferably selected from G, A or V, X3, X6, X8, X10, X11, X12, X14 and X15 are negatively charged amino acids, preferably selected from E or D, X4 is an amino acid with a polar uncharged side chain, preferably selected from S, T, N or Q, and X16 is a large hydrophobic amino acid, selected from Y or F.
- (17) For instance, the peptidic moiety of the peptidic based inhibitor candidate has the amino acid sequence selected from Y, EAY, EDY and EEY.
- (18) In a particular embodiment, the peptidic based inhibitor candidate further comprises a reactive moiety, preferably selected from epoxysuccinyl, acyloxymethyl, aldehydes and ketones. In an embodiment, the reactive group is incorporated within the peptidic sequence. For instance, the reactive group is an epoxyde, which replaces the glutamate residue adjacent to the very C-terminal aromatic residue, preferably F or Y.
- (19) It is another object of the invention to provide a peptidic based inhibitor for use in the treatment of a disorder involving altered microtubule detyrosination in an animal, wherein the peptidic based inhibitor comprises or a peptidic moiety constituted of 1 to 20 amino acids, said peptidic moiety having an amino acid selected from Y or F at the C-terminal position, and wherein the peptidic based inhibitor inhibits at least partially a tubulin carboxypeptidase activity.
- (20) According to the invention, the peptidic based inhibitor inhibits irreversibly or reversibly a tubulin carboxypeptidase activity.
- (21) The disorder is preferably selected from neurodegenerative diseases, preferably selected from Alzheimer disease, Parkinson disease, psychiatric disorders, and neural disorders, neuronal regeneration disorders, cancers, preferably selected from colon cancer and neuroblastoma, muscular dystrophies, heart diseases, vascular disorders, infertility, retinal degeneration, and ciliopathies.
- (22) It is another object of the present invention to provide a pharmaceutical composition comprising a therapeutically effective amount of such peptidic based inhibitors.
-

Description

FIGURES

- (1) FIG. 1: schematic overview of detyrosination and tyrosination cycle of microtubule, which consists of the removal of the very C-terminal tyrosine from α -tubulin by use of Tubulin CarboxyPeptidase (TCP) and results in generation of so-called $\Delta 1$ -tubulin. Incorporation of tyrosine (Y) at the very C-terminus of the detyrosinated soluble tubulin is obtained by Tubulin tyrosine ligase (TTL).
- (2) FIG. 2: isolation of native TCP activity from a brain extract by way of cycles of depolymerization/polymerization to isolate and purify Microtubule Associated Proteins (MAPs).
- (3) FIG. 3: biochemical isolation of MAPs before (MAPs) and after (Enriched fraction) enrichment by biochemical enrichment including ammonium sulphate precipitation and hydrophobic chromatography. (The polymerization/depolymerization method is used to get the initial MAPs. The enriched fraction is after the biochemical procedure involving ammonium sulphate and hydrophobic chromatography.)
- (4) FIG. 4: (A) schematic representation of detyrosination assay wherein .sup.3H Tyrosine is incorporated

into soluble Tubulin by use of TTL. Following polymerization cycles .sup.3H Tyr-Tubulin is incorporated in microtubules to obtain .sup.3H Tyr-microtubule; (B) validation of the TCP activity contained in the isolated MAPs fraction.

(5) FIG. 5: inhibition of native TCP activity in brain extract by treatment with different Aspartic, Cysteine, Metallo and Serine proteases inhibitors (ASP=aspartic protease inhibitor, CYS=cysteine protease inhibitors, METALLO=metalloprotease inhibitors, SER=serine protease inhibitors).

(6) FIG. 6: Immunoblot analysis of protein extract obtained from HEK293 ectopically expressing individual CRMP family members. Δ1-tubulin represents the amounts of detyrosinated tubulin. HA displays the level of ectopically expressed CRMP members. The α-tubulin labeling served as a loading control and allows to compare the ratio of detyrosinated tubulin to total tubulin.

(7) FIG. 7: Immunofluorescence analysis of U2OS cells ectopically expressing the five members of CRMP family. The left panel displays the immunofluorescence (IF) signal for the ectopically expressed CRMPs. In the right panel the signal can be observed for detyrosinated MTs.

(8) FIG. 8: Immunoblot showing specific knockdown of endogenous CRMP1 expression in U2OS cells by siRNA interference. The α-tubulin labeling serves as a loading control and allows to compare the ratio of detyrosinated tubulin to total tubulin.

(9) FIG. 9: Immunofluorescence analysis of U2OS cells depleted for CRMP1 protein. The left panel displays the total tubulin level per cell. On the right panel only microtubules labeled for detyrosination are staining positively.

(10) FIG. 10: (A) Immunoblot analysis of U2OS cells depleted for CRMP1 protein. Cell cycle arrest is demonstrated by accumulation of cyclin dependent kinase inhibitor p21 and its downstream effector p53 as well as decrease in the amount of phosphorylated histone 3 (serine—Ser10P). (B) Flow cytometry analysis of bulk DNA content of CRMP1 depleted U2OS cells. (C) Graphical representation of the relative number of cells in different cell cycle stages from control (Luciferase) and CRMP1 depleted U2OS cells.

(11) FIG. 11: (A) inhibition of native TCP activity in brain-derived MAPs resulting from incubation with various peptides inspired by the C-terminal sequence of tubulin (EDY, EEY). (B) Dose response curve of TCP activity in presence of increasing EEY concentrations.

(12) FIG. 12: C2C12 muscle differentiation model. Time-course of C2C12 cells mimicking muscle differentiation in presence or absence of the TCP inhibitor EEY. Immunoblotting analysis of protein extracts obtained from myogenic differentiation of C2C12 cells. Incubation with EEY led to decrease tubulin detyrosination levels (Δ1-tubulin).

(13) FIG. 13: SH-SY5Y neural differentiation process. (A) Phase contrast microscopy pictures of SH-SY5Y cells at Day 0 and Day 8 after neural differentiation. (B) Gene expression analysis of DDC (Aromatic-L-amino-acid decarboxylase), a marker of dopaminergic neurons and of CRMP1 (C1) expression during the neuronal differentiation process.

(14) FIG. 14: Immunoblot analysis of detyrosinated tubulin (deTyr-tub) of Control and DMD cells that contain a causal genetic mutation for Duchenne Muscular Dystrophy.

(15) FIG. 15: Immunoblot analysis of detyrosinated tubulin (deTyr-tub) of Control and SH-SY5Y cells.

(16) FIG. 16: Immunoblot analysis of detyrosinated tubulin (deTyr-tub) of Control and iPSC generated from skin fibroblasts from patients carrying two different familial Alzheimer's Disease mutations (Mut_1 and Mut_2).

(17) FIG. 17: Immunoblot analysis of detyrosinated tubulin (deTyr-tub) of Control and CHL-1 cells and HEK cells, in presence or absence of TCPase inhibitor (Eps-Y).

(18) FIG. 18: Examples of peptide-based inhibitors comprising a peptidic moiety and a reactive group composed of, for example, an epoxyde group. A first general example illustrates a subtype of different inhibitors composed of a very C-terminal tyrosine (Y) attached to an epoxyde group and the C-terminal sequence of alpha tubulin such as GEepoxydeY (inh1). In the chemical formula, R1 represents the amino acid sequence of human alpha tubulin and R2 a wide variety of C-terminal modifications such as, but not limited to, COOH; CONH2, NH2, aldehyde, pNA, Amc, hydrazide, hydroxamic acid, CMK). Those modifications may contribute to preventing enzyme degradation, to mimic native proteins, and in some cases to remove hydrogen bonding at the C-terminal of the peptides, tools for studying structure-activity relationship (SAR), and more. The second molecule (inh2) represent the formula of a shorter version of

the peptidic inhibitor.

DETAILED DESCRIPTION OF THE INVENTION

(19) The present invention originally proposes to exploit the natural substrate of proteins with TCP activity, i.e., the very C-terminal sequence of α -tubulin, as pharmacological tool for inhibiting TCP activity. The inventors discovered several families of proteins possessing TCP activity in a controlled purified biochemical assay as well as in human cell cultures. In turn, the inventors have developed a method to design pharmacological compounds that specifically inhibit (either partially or irreversibly) detyrosinase activity in cellulo. A plethora of compounds that selectively act on TCP activity can be designed according to the invention, for applications as research tools and most promisingly in treatment for several disorders such as neurodegenerative diseases and psychiatric disorders.

(20) Method for Purifying Proteins Having a Tubulin Carboxypeptidase Activity

(21) The present invention proposes a method suitable for purifying proteins having a tubulin carboxypeptidase activity from a biological extract.

(22) In the context of the present invention, the terms “protein having a tubulin carboxypeptidase activity” or “protein having a TCP activity” or “TCPase protein” or “TCP” are used for referring to a class of proteins that are able to cleave off the Glu-Tyr bond to release the C-terminal tyrosine residue from a native tyrosinated tubulin (FIG. 1).

(23) The term “biological sample” means any sample derived from an animal, including multi- or unicellular organisms, which contains microtubules. Preferably, the biological sample derived from a mammal, preferably selected from pig, monkey, human, rat or mouse. Examples of such biological samples include fluids, tissues, cell samples, organs, biopsies, etc. Most preferred samples are brain extract, testis extract, and lung extract.

(24) The biological sample may be treated prior to its use, e.g. in order to render the microtubules available. Techniques of cell lysis, concentration or dilution of microtubules, are known by the skilled person.

(25) According to the invention, the method for purifying proteins having a tubulin carboxypeptidase activity from a biological extract, comprises: (a) centrifuging the biological extract at a temperature comprised between 0 and 10° C., preferably between 2 and 5° C., more preferably at 2° C.; (b) recovering the supernatant from step (a) and proceeding to a first microtubule polymerization cycle by adding GTP and incubating the mixture at a temperature between 35 and 40° C., preferably at 37° C., $\pm 2^\circ$ C., then centrifuging; (c) recovering the pellets of step (b), resuspending in ice-cold buffer, incubating at 4° C., $\pm 1^\circ$ C., and proceeding to a second microtubule polymerization cycle by adding GTP and incubating the mixture at 37° C., $\pm 2^\circ$ C., then centrifuging; (d) recovering the pellets of step (c) resuspending in ice-cold buffer, incubating at 4° C. $\pm 1^\circ$ C., and proceeding to a third microtubule polymerization cycle by adding GTP and incubating the mixture at 37° C., $\pm 2^\circ$ C., then centrifuging; (e) resuspending the pellets of step (d) and submitting the mixture to an ion exchange chromatography and recovering the flow through; (f) precipitating the proteins of the flow through with a 60% saturated ammonium sulphate solution; (g) submitting the precipitated fraction of step (f) to an hydrophobic chromatography and eluting by gradually decreasing ammonium sulphate concentration up to zero to recover the fraction of proteins with a tubulin carboxypeptidase activity.

(26) In a particular embodiment, the first polymerization cycle comprises (i) adding GTP and incubating the mixture at 37° C., $\pm 2^\circ$ C., for 30 minutes, ± 10 minutes; (ii) centrifuging at 22,000 g, $\pm 1,000$ g, at 37° C., $\pm 2^\circ$ C., for 45 minutes, ± 10 minutes.

(27) Alternatively or in addition the second polymerization cycle may comprise (i) incubating the mixture on ice for 30 minutes, ± 10 minutes; (ii) centrifuging at 150,000 g $\pm 10,000$ g, 30 minutes, ± 10 minutes; (iii) recovering the supernatant and adding GTP; (iv) incubating the mixture at 37° C., $\pm 2^\circ$ C., for at 30 minutes, ± 10 minutes; (v) centrifuging at 50,000 g, $\pm 1,000$ g at a temperature comprised between 30° C. and 37° C., for 30 minutes, ± 10 minutes.

(28) Alternatively or in addition the third polymerization cycle may comprise (i) incubating the mixture on ice for 30 minutes, ± 10 minutes; (ii) centrifuging at 150,000 g, $\pm 10,000$ g, 30 minutes, ± 10 minutes; (iii) recovering the supernatant and adding GTP; (iv) incubating the mixture at 37° C., $\pm 2^\circ$ C., for at 30 minutes, ± 10 minutes; (v) centrifuging at 50,000 g, $\pm 1,000$ g at a temperature comprised between 30° C. and 37° C., for 30 minutes, ± 10 minutes.

- (29) In a particular embodiment, the method further comprises a step of mass spectrometry characterization of the fraction of proteins of step (g).
- (30) In a particular embodiment, the method further comprises a step of selecting proteins that contain a protease domain. To determine what type of protease activity would be required for detyrosination, various inhibitors of cysteine, aspartic, serine, threonine proteases and metalloproteases may be tested.
- (31) In a particular embodiment, the fraction of proteins with a tubulin carboxypeptidase activity comprises at least one protein having at least 30%, 40%, 50%, 60%, 70%, 80%, 90%, 95%, 99% amino acid sequence identity with the amino acid sequence selected from SEQ ID NO: 1, SEQ ID NO: 2, SEQ ID NO: 3, SEQ ID NO: 4, SEQ ID NO: 5, SEQ ID NO: 6, SEQ ID NO: 7, SEQ ID NO: 8, SEQ ID NO: 9, SEQ ID NO: 10 and SEQ ID NO: 11.
- (32) In a particular embodiment, the fraction of proteins with a tubulin carboxypeptidase activity is obtained from a brain extract, such as a brain extract from pigs, and the mass spectrometric data are aligned with human reference sequences, in order to identify corresponding human proteins.
- (33) Advantageously, such fraction of proteins comprises at least one protein selected from human Ubiquitin carboxyl-terminal hydrolase 14 (UBP14—SEQ ID NO: 1), human Ubiquitin carboxyl-terminal hydrolase 5 (UBP5—SEQ ID NO: 2), human Methionine aminopeptidase 2 (MAP2—SEQ ID NO: 3), human Xaa-Pro aminopeptidase 1 (XPP1—SEQ ID NO: 4), human Tripeptidyl-peptidase 2 (TPP2—SEQ ID NO: 5), human Vasohibin-1 (VASH1—SEQ ID NO: 6), human dihydropyrimidinase-related protein 1 (DPYL1—SEQ ID NO: 7), human dihydropyrimidinase-related protein 2 (DPYL2—SEQ ID NO: 8), human dihydropyrimidinase-related protein 3 (DPYL3—SEQ ID NO: 9), human dihydropyrimidinase-related protein 4 (DPYL4—SEQ ID NO: 10) and human dihydropyrimidinase-related protein 5 (DPYL5—SEQ ID NO: 11).
- (34) In another embodiment, the fraction of proteins comprises at least one protein selected from the proteins listed in Table 1.
- (35) Advantageously, the fraction of proteins with a tubulin carboxypeptidase activity is further contacted with microtubules and the level of isolated tyrosine (Y) is measured, thereby confirming the tubulin carboxypeptidase activity of the fraction of proteins. For instance, the microtubules comprise synthetic microtubules and/or α -tubulins, with labeled C-terminal Y.
- (36) After a time of contact of the fraction of proteins with putative TCP activity with microtubules, under conditions suitable for the proteins with putative TCP activity to perform detyrosination of the microtubules and/or α -tubulin, the amount of isolated Y, or free Y, in the sample is measured. For instance, tubulin tyrosine ligase (TTL) enzyme incorporates radioactively labeled .sup.3H-tyrosine at the very C-terminus of detyrosinated soluble tubulin obtained from e.g. brain extracts. The radioactively labeled tubulin is incorporated in MT during a polymerization cycle to obtain radioactively labeled MTs. TCP activity contained by the biological sample will cut off the radioactively marked tyrosine which can be quantified by scintillation analysis. Alternatively, TCP activity can be monitored by comparing the ratio of detyrosinated versus tyrosinated tubulin before and after exposure to the biological sample by immunoblot analysis.
- (37) Several factors may affect the rate at which enzymatic reactions proceed: temperature, pH, enzyme concentration, substrate concentration, and the presence of any inhibitors or activators.
- (38) In some embodiments, it is possible to employ a buffer containing a nucleoside triphosphate, such as ATP, potassium chloride, magnesium chloride, and a reducing agent such as DTT in order to provide optimal conditions for the enzymes with putative TCP activity to detyrosinate the microtubules and/or α -tubulin.
- (39) The pH value is preferably in the range of 5 to 9, in order to provide suitable conditions for the enzymes with putative TCP activity to detyrosinate the microtubules and/or α -tubulin. More preferably, the pH value is between 5.5 and 8.5, even more preferably between 6 and 8.
- (40) A suitable reaction time for enzymes with putative TCP activity to detyrosinate the microtubules and/or α -tubulin may be in the range of 5 minutes to 10 hours, preferably 10 minutes to 5 hours, more preferably 1 hour to 3 hours.
- (41) In a particular embodiment, the concentration of fraction of proteins with a tubulin carboxypeptidase activity contacted with microtubules and/or α -tubulin is in the range of 0.1 μ M to 1 mM, preferably 0.25 μ M to 500 μ M, more preferably 0.5 μ M to 300 μ M, and even more preferably 1 μ M to 200 μ M, in order

to provide optimal conditions for detyrosination of the microtubules and/or α -tubulin.

(42) In a particular embodiment, the fraction of proteins with a tubulin carboxypeptidase activity is contacted with at least α -tubulin. In another particular embodiment, the fraction of proteins with a tubulin carboxypeptidase activity is contacted with a polypeptide corresponding to the C-terminus of α -tubulin. In another embodiment, the fraction of proteins with a tubulin carboxypeptidase activity is contacted with a mixture of microtubules and α -tubulin.

(43) Advantageously, the microtubules and/or α -tubulin comprise synthetic microtubules/peptides and/or α -tubulins, wherein the α -tubulin comprises labeled C-terminal Y, so that step of measuring free Y may be easily implemented.

(44) According to the invention, the tubulin carboxypeptidase activity of the fraction of proteins is confirmed if detectable isolated Y in the sample/microtubules and/or α -tubulin in the sample is observed and compared to a fraction of protein that lacks TCP activity (negative control).

(45) By “synthetic microtubules/peptides and/or α -tubulin” it is intended a microtubule or α -tubulin that has been chemically constructed. The synthetic microtubules or α -tubulin may be artificially constructed by methods of synthetic biology, including solid phase peptide synthesis (SPPS), prior thiol capture strategy, native chemical ligation (NCL). The term “Synthetic microtubules and/or α -tubulin” also encompasses natural microtubule or α -tubulin that has been treated to change its C-terminal amino acid by a labeled -Y.

(46) According to the invention, the labeled -Y consists of a tyrosine that is labeled with a molecule or material that can produce a detectable (such as visually, electronically, radioactively, or otherwise) signal that indicates the presence and/or concentration of the tyrosine in a sample. Thereby, e.g., the presence, location and/or concentration of the tyrosine in a sample can be detected by detecting the signal produced by the detectable molecule or material. The labeled -Y can be detected directly or indirectly. In certain embodiments, the label, or detectable molecule or material, may react with a suitable substrate (e.g., a luciferin) to generate a detectable signal. In particular, the detectable label can be a fluorophore, an enzyme (peroxidase, luciferase), a radioisotope, a fluorescent protein, or a fluorescent dye. Other detectable molecule or material including chemiluminescent labels, electrochemiluminescent labels, bioluminescent labels, polymers, polymer particles, metal particles, haptens, and dyes.

(47) In a particular embodiment, the level of isolated Y in the sample is compared to the level of isolated Y in a control sample comprising solely microtubules and/or α -tubulin. The control sample is free of the fraction of proteins with a tubulin carboxypeptidase activity, so that the difference between both amounts of isolated Y can be attributed to said compound.

(48) Peptidic Based Inhibitors Able to Inhibit Tubulin Carboxypeptidase Activity

(49) According to the invention, the peptidic based inhibitors block or reduce the tubulin carboxypeptidase activity of an enzyme. A peptidic inhibitor can act with competitive, uncompetitive or noncompetitive inhibition. A peptidic inhibitor of the invention can bind reversibly or irreversibly, and therefore the term includes compounds that are suicide substrates of an enzyme with TCP activity.

(50) The present invention relates to a method for selecting a peptidic based inhibitor able to inhibit a tubulin carboxypeptidase activity that has been selected and designed based on the natural C-terminal sequence of α -tubulin.

(51) More particularly, the inventors have developed a method, wherein a peptidic inhibitor containing a peptidic moiety constituted of 1 to 20 amino acids, wherein the most C-terminal amino acid is selected from Y or F, is contacted with a biological extract from an animal, in order to identify and isolate a peptide having a tubulin carboxypeptidase activity.

(52) Therefore, it is an object of the invention to provide a method for selecting a peptidic based inhibitor able to inhibit a tubulin carboxypeptidase activity among peptidic based inhibitor candidates that comprise a peptidic moiety constituted of 1 to 20 amino acids, said peptidic moiety having at the C-terminal position an amino acid selected from Y or F, wherein the method comprises a step (a) of contacting the peptidic based inhibitor candidate with a mixture containing both a fraction of protein with a tubulin carboxypeptidase activity and microtubules; and a step (b) of measuring the level of isolated Y and/or detyrosinated microtubules.

(53) As used herein, the terms “tubulin carboxypeptidase inhibitor” or “peptidic based inhibitor” refers to a class of molecules that target and inhibit, at least partially, the activity of proteins having a tubulin

carboxypeptidase activity, and thereby inhibit microtubule detyrosination.

(54) The amino acid sequences defined herein use the one letter code as following: A: Ala (alanine); R: Arg (arginine); N: Asn (asparagine); D: Asp (aspartic acid); C: Cys (cysteine); Q: Gln (glutamine); E: Glu (glutamic acid); G: Gly (glycine); H: His (histidine); I: Ile (isoleucine); L: Leu (leucine); K: Lys (lysine); M: Met (methionine); F: Phe (phenylalanine); P: Pro (proline); S: Ser (serine); T: Thr (threonine); W: Trp (tryptophan); Y: Tyr (tyrosine); V: Val (valine).

(55) The amino acid sequences may also comprise non-naturally-occurring amino acid such as azetidinecarboxylic acid, 2-aminoadipic acid, 3-aminoadipic acid, beta-alanine, aminopropionic acid, 2-aminobutyric acid, A-aminobutyric acid, 6-aminocaproic acid, 2-aminoheptanoic acid, 2-aminoisobutyric acid, 3-aminoisobutyric acid, 2-aminopimelic acid, 2,4 diaminoisobutyric acid, desmosine, 2,2'-diaminopimelic acid, 2,3-diaminopropionic acid, N-ethylglycine, N-ethylasparagine, hydroxylysine, allo-hydroxylysine, 3-hydroxyproline, 4-hydroxyproline, isodesmosine, allo-isoleucine, N-methylglycine, N-methylisoleucine, N-methylvaline, norvaline, norleucine, ornithine, selenocysteine, nitrotyrosine, dihydroxyphenylalanine, and pipecolic acid.

(56) The term "peptide" refers herein to a polymer of amino acid residues linked together by peptide (amide) bonds. Said term also encompasses fragments of polypeptides. Said fragments have preferably biological activity. Said fragments may have a length of 10, 20, 30, 40, 50, 60, 70, 80, 90, 100, 150, 200, 250, 300, 350, 400, 450 or more amino acids.

(57) The term "peptidic moiety" refers to a moiety containing at least one amino acid and at most 20 amino acids. When the peptidic moiety comprises two or more amino acids, said amino acids are linked together by peptide bonds and chemically modified or not.

(58) According to the method of the invention for selecting a peptidic based inhibitor able to inhibit a tubulin carboxypeptidase activity, the peptidic based inhibitor candidate is contacted with a mixture containing both a fraction of protein with a tubulin carboxypeptidase activity and microtubules (step a), and the rate of inhibition of TCP activity is calculated by measuring the level of isolated Y and/or deetyrosinated microtubules (step b).

(59) In some embodiments, said method for selecting a peptidic based inhibitor able to inhibit a tubulin carboxypeptidase activity among peptidic based inhibitor candidates that comprise a peptidic moiety constituted of 1 to 20 amino acids, said peptidic moiety having at the C-terminal position an amino acid selected from Y or F, wherein the method comprises: (a) contacting the peptidic based inhibitor candidate with a mixture containing both a fraction of native or recombinant proteins with a tubulin carboxypeptidase activity and microtubules, which preferably comprise synthetic microtubules and/or α -tubulins, with labeled C-terminal Y; (b) measuring the level of isolated Y and/or deetyrosinated microtubules.

(60) In a particular embodiment, the reaction temperature is maintained in the range of 1° C. to 70° C., preferably 5° C. to 65° C., more preferably 10° C. to 60° C., even more preferably 15° C. to 55° C., most preferably 19° C. to 43° C., and for example 19° C. to 37° C. in order to provide optimal conditions for the putative TCPase enzyme to deetyrosinate the microtubules and/or α -tubulin.

(61) The method of the invention may be implemented with a large kind of peptidic based inhibitors that share sequence identity or homology with the C-terminal amino acid sequence of α -tubulin.

(62) More particularly, the peptidic based inhibitor of the invention comprises a peptidic moiety constituted of 1 to 20 amino acids, said peptidic moiety having at the C-terminal position an amino acid selected from Y or F.

(63) According to the invention, in the three-dimensional conformation of the peptidic moiety the C-terminal Y or F is accessible to enzymes, and more particularly to proteins having a TCP activity.

(64) In a preferred embodiment, the peptidic moiety is constituted of the 1 to 20 amino acid of the most C-terminal amino acid part of alpha-tubulin.

(65) In a particular embodiment, the peptidic moiety is constituted of between 1 and 16 of the most C-terminal amino acids of the amino acid sequence Nter-X1-X2-X3-X4-X5-X6-X7-X8-X9-X10-X11-X12-X13-X14-X15-X16-Cter (SEQ ID NO: 20), wherein X1, X2, X5, X7, X9 and X13 are hydrophobic amino acids, preferably selected from G, A or V, X3, X6, X8, X10, X11, X12, X14 and X15 are negatively charged amino acids, preferably selected from E or D, X4 is an amino acid with a polar uncharged side chain, preferably selected from S, T, N or Q, and X16 is a large hydrophobic amino acid,

preferably selected from Y or F.

(66) In general, "X" can denote any amino acid unless indicated otherwise herein.

(67) The physicochemical groups are generally defined as following: the non-polar or hydrophobic amino acids including A, V, I, L, P, F, M, and W, but more narrowly the non-aromatic hydrophobic amino acids as including A, V, I, L, P, and M; the uncharged polar group including G, S, T, C, Y, N and Q; the negatively charged polar group including E and D; and the positively charged polar group including R and K.

(68) X16 refers to the ultimate C-terminal amino acid in the peptidic moiety. Preferably, X16 is Y. The other amino acids are optional. The peptidic moiety may comprise all or part of the amino acids of SEQ ID NO: 12, with respect of the numeration, wherein X1, if present is the N-terminal amino acid in the peptidic moiety, and so on.

(69) In a particular embodiment, the amino acid sequence of the peptidic moiety consists on Y (X16).

(70) In another particular embodiment, the amino acid sequence of the peptidic moiety consists on EDY.

(71) In another particular embodiment, the amino acid sequence of the peptidic moiety consists on EEY.

(72) In another particular embodiment, the amino acid sequence of the peptidic moiety consists on EAY.

(73) In another embodiment, the amino acid sequence of the peptidic moiety comprises or consists on the amino acid sequence selected from SEQ ID NO: 13, SEQ ID NO: 14, SEQ ID NO: 15, SEQ ID NO: 16, SEQ ID NO: 17, SEQ ID NO: 18.

(74) The peptidic inhibitor may inhibit irreversibly or reversibly a tubulin carboxypeptidase activity. As an example, a chemically modified amino acid that may irreversibly react with the cysteine contained by the protein with TCP activity, such as a catalytic triad, is considered to be irreversible. On the other hand a peptide, or chemically modified peptide, that does not covalently react or form reversible bonds with the thiol group contained by the enzyme may be washed off and is considered as reversible.

Iodoacetamide is an irreversible inhibitor of all cysteine peptidases, with the mechanism of inhibition occurring from alkylation of the catalytic cysteine residue.

(75) According to the invention, the activity of the peptidic inhibitor may be modulated by expanding the number of amino acid residues and/or by use of a reactive moiety, preferably selected from epoxysuccinyl (Eps), acyloxymethyl, aldehydes and ketones. Such reactive moiety that further functionalizes the peptide may be linked to the peptidic inhibitor by use of known methods in the art, such as, example given, methods of synthetic medicinal chemistry, synthesis of various intermediates, deuterated forms of the compounds and stereoisomers thereof (FIG. 18).

(76) For instance, the peptidic inhibitors is Eps-EEY. Such peptidic inhibitor act as a reversible inhibitor of TCPase activity.

(77) The present invention thus provides a method to design plethora of peptidic inhibitors able to inhibit a tubulin carboxypeptidase activity.

(78) Therapeutic Use of Peptidic Based Inhibitors

(79) As well exposed above, detyrosination of microtubules is associated with cancer progression, aberrant neuronal networks, weak neuronal remodeling, plasticity and/or adaptation. Accordingly, the use of such peptidic inhibitors may have a positive impact in the treatment of disorders involving microtubule detyrosination. For instance, peptidic inhibitors of the present invention may be used for increasing the microtubule dynamics and thereby impacting neuroregeneration.

(80) The present invention thus relates to peptidic based inhibitor for use in the treatment of heart disorder, vascular disorder, cancers, neurodegenerative disorders, muscle disorders, infertility, ciliopathies, more generally a disorder involving altered microtubule detyrosination in an animal, preferably but not limited to a mammal, wherein the peptidic based inhibitor comprises a peptidic moiety constituted of 1 to 20 amino acids, said peptidic moiety having an amino acid selected from Y or F at the C-terminal position, and wherein the peptidic based inhibitor inhibits at least partially a tubulin carboxypeptidase activity. In a particular embodiment, the peptidic based inhibitor comprises GVDSVEAEAEEGEEY (SEQ ID NO: 19). In another embodiment, the peptidic based inhibitor comprises GEEY.

(81) Thus, peptidic inhibitors of the present invention are good candidate for treating neurodegenerative diseases, preferably selected from Alzheimer disease, Parkinson disease, psychiatric disorders, and neural disorders, neuronal regeneration disorders, cancers, preferably selected from colon cancer and

neuroblastoma, muscle disorders such as muscular dystrophies, retinal degeneration, heart diseases, vascular disorders, infertility, and ciliopathies.

(82) It is an object of the present invention to provide peptidic based inhibitor for use for treating neurodegenerative diseases. In a particular embodiment, the neurodegenerative disease is Alzheimer disease.

(83) It is a further embodiment to provide peptidic based inhibitor for use for treating cancers.

(84) It is a further embodiment to provide peptidic based inhibitor for use for treating muscular dystrophies, particularly Duchenne muscular dystrophy.

(85) The invention additionally provides a pharmaceutical composition comprising a therapeutically effective amount of a peptidic inhibitors according of the invention.

(86) By “therapeutically effective amount” is meant an amount of the peptidic inhibitor of the invention that elicits a desired therapeutic effect. The exact amount dose will depend on the purpose of the treatment and will be ascertainable by one skilled in the art using known techniques. As is known in the art, adjustments for age, body weight, general health, sex, diet, drug interaction and the severity of the condition may be necessary, and will be ascertainable with routine experimentation by those skilled in the art.

(87) A pharmaceutical composition according to the present invention may further comprise one or more pharmaceutically acceptable carriers. In a specific embodiment, the term “pharmaceutically acceptable” means approved by a regulatory agency or other generally recognized pharmacopoeia for use in animals, and more particularly in humans.

(88) Pharmaceutically acceptable carriers are well known in the art and include, for example, aqueous solutions such as water, 5% dextrose, or physiologically buffered saline or other solvents or vehicles such as glycols, glycerol, oils such as olive oil, or injectable organic esters that are suitable for administration to a human or non-human subject. In some embodiments, a pharmaceutically acceptable carrier or composition is sterile. A pharmaceutical composition can comprise, in addition to the active agent, physiologically acceptable compounds that act, for example, as bulking agents, fillers, solubilizers, stabilizers, osmotic agents, uptake enhancers, etc. Physiologically acceptable compounds include, for example, carbohydrates, such as glucose, sucrose, lactose, dextrans, polyols such as mannitol, antioxidants, such as ascorbic acid or glutathione, preservatives, chelating agents, buffers, or other stabilizers or excipients.

(89) The choice of a pharmaceutically acceptable carrier(s) and/or physiologically acceptable compound(s) can depend for example, on the nature of the active agent, e.g., solubility, compatibility (meaning that the substances can be present together in the composition without interacting in a manner that would substantially reduce the pharmaceutical efficacy of the pharmaceutical composition under ordinary use situations) and/or route of administration of the composition.

(90) Pharmaceutical compositions of the invention comprise a therapeutically effective amount of one or several peptidic inhibitors according to the invention and can be formulated in various forms, e.g. in solid, liquid, gaseous or lyophilized form and may be, inter alia, in the form of an ointment, a cream, transdermal patches, a gel, powder, a tablet, solution, an aerosol, granules, pills, suspensions, emulsions, capsules, syrups, liquids, elixirs, extracts, tincture or fluid extracts or in a form which is particularly suitable for topical or oral administration. A variety of routes are applicable for administration of the polypeptide of the invention, including, but not limited to, orally, topically, transdermal, subcutaneously, intravenously, intraperitoneally, intramuscularly or intraocularly. However, any other route may readily be chosen by the person skilled in the art if desired.

(91) The pharmaceutical compositions can be used for the treatment of a wide variety of different diseases and disorders. Thus the invention also encompasses methods of treatment comprising administering a therapeutically effective amount of a peptidic inhibitors of the invention to a subject in need thereof. The subject is typically a mammal, e.g., a human. In some embodiments the subject is a non-human animal that serves as a model for a disease or disorder that affects humans. The animal model may be used, e.g., in preclinical studies, e.g., to assess efficacy and/or determine a suitable dose.

(92) In some embodiments, a peptidic inhibitor of the invention is administered prophylactically, e.g., to a subject who does not exhibit signs or symptoms of the disease or disorder (but may be at increased risk of developing the disorder or is expected to develop the disease or disorder).

EXAMPLES

Example 1: Isolation of Native TCP Activity from Brain Extract

(93) The method for purifying proteins having a tubulin carboxypeptidase activity of the invention has been performed on porcine brain extract. More particularly, microtubule associated proteins (MAPs) were isolated from the crude brain extract (FIG. 2) as exposed below.

Material & Method

(94) PEM-Buffer Composition:

(95) TABLE-US-00001 concentration chemical stock 50 mM PIPES/NaOH, pH 6.8 400 mM, 4° C. 1 mM EGTA 100 mM, RT 1 mM MgCl₂ 1 M, RT

Experimental Procedure

(96) Pig brains were quickly removed from the skull and cooled down in ice-cold water by shaking and they were kept on ice. For each 10 g of brain material, 15 ml of PEM buffer containing 1 μ l β -mercaptoethanol were added. The brains were pre-homogenised in a mixer and then transferred into a Potter homogeniser (on ice). The extract was spun for 1 h at 22,000 g at 2° C. and the supernatant was removed carefully.

(97) 1st Polymerization Cycle:

(98) The supernatant was supplemented with 1 mM GTP and incubated at 37° C. for 30 minutes while the solution was stirred smoothly. Following the incubation, sample was spun down at 22,000 g for 45 min. The pellet containing microtubules and Microtubule Associated Proteins (MAPs) was kept while the supernatant was discarded.

(99) 2^{sup}.nd and 3rd Polymerization Cycle:

(100) The pellet was re-suspended in 0.1 vol. of the initial volume in ice-cold PEM buffer containing 0.1 mM GTP and re-homogenised in an ice-cold Potter homogeniser. Next, the suspension was incubated on ice for 30 min and centrifuged for 30 min at 150,000 g at 2° C. (41,000 rpm in a 50-2Ti rotor). The pellet was discarded.

(101) The supernatant was adjusted to 1 mM GTP and incubated in pre-weighted centrifuge tubes for 30 min at 37° C. and after spun for 30 min at 30-37° C., 50,000 g (24,000 rpm in a 50-2Ti rotor). The supernatant was discarded while the pellet, which contains microtubules and MAPs was re-suspended in PEM buffer supplemented with 1 mM GTP and subjected to 3^{sup}.rd polymerization cycle.

(102) Following the third cycle the pellet containing microtubules and MAPs was re-suspended and subjected to DEAE-Sephadex ion exchange chromatography. This step serves to separate the tubulin (microtubules), which was found associated with the column while MAPs containing the TCP activity were found in the flow through.

(103) The flow through from the DEAE column was collected and the MAPs were concentrated. Differential ammonium sulphate precipitation was used. The majority of the TCP activity was recovered at 60% of saturated ammonium sulphate solution. This is a critical step experimentally obtained to yield a MAP fraction containing native TCP activity.

(104) To further enrich for TCP activity an additional inventive step consisting of chromatography based on hydrophobic interactions was added. The MAP fraction recovered from ammonium sulphate precipitation was loaded on phenyl sepharose chromatography and eluted by gradually decreasing ammonium sulphate concentration in order to optimize the recovery of native TCP activity from brain extracts. The presence of ammonium sulphate strongly increases surface tension in aqueous solutions and promotes hydrophobic interactions.

(105) Following the elution, the fraction with the highest TCP activity (FIG. 3) was subjected to characterization by mass spectrometry. Importantly the fraction was analyzed in a dephosphorylation assay to confirm the presence of TCPase activity (FIG. 4). Mass spectrometry analysis of the enriched fraction (FIG. 3) yielded a total of 584 identified proteins, as listed in Table 1 below, which were analyzed by functional homology search to specifically identify TCP candidates based on the presence of protease domain. TCP candidates are peptidases, enzymes that hydrolysis peptide bonds. In the context of the invention, proteases, proteinases and proteolytic enzymes are used interchangeably. To search for potential candidates that contain a protease domain, the obtained list of peptides was analysed for Conserved Domain. This was performed using Conserved Domain Database (CDD) a curated database that annotates functional units in proteins (hosted by NCBI). The collection of domain models includes a

set curated by NCBI, which utilizes 3D structure to provide insights into sequence/structure/function relationships. In as such to shorten the list of proteins obtained after specific enrichment down to potential candidates having at least one potential protease domain, enquiries were performed. Besides, further description of the candidates was obtained using MEROPS database a second independent resource for information on peptidases (merops.sanger.ac.uk/about/index.shtml). Additionally, proteins with newly identified protease domain were also screened in literature. This resulted in a selection of 11 potential TCP candidates (SEQ ID NO: 1 to SEQ ID NO: 11), among which the family of proteins consisted of collapsin response mediator protein (CRMPs) has been characterized as example.

(106) TABLE-US-00002
TABLE 1 List of microtubules associated proteins obtained after purification steps (>2 peptides per hit)
Mol Sequence UNIPROT weight coverage ID PROTEINS GENES [kDa] [%]
A2AGT5 Cytoskeleton-associated protein 5 Ckap5 226 16.3 A2AJI0 MAP7 domain-containing protein 1 Map7d1 93 3.4 D3Z2H9 Uncharacterized protein Tpm3-rs7 29 23.8 E9PY16 ArfGAP with dual PH domains 1 Adap1 43 15.8 E9Q557 Desmoplakin Dsp 333 0.7 E9Q912 RAP1, GTP-GDP dissociation stimulator 1 Rap1gds1 66 15.3 E9QAS7 Inositol polyphosphate-5-phosphatase A Inpp5a 49 14.5 F8VPN4 Amylo-1,6-glucosidase, 4-alpha-glucanotransferase Agl 174 1.4 O08532 Voltage-dependent calcium channel subunit alpha- Cacna2d1 125 3.4 2/delta-1 O08539 Myc box-dependent-interacting protein 1 Bin1 64 5.4 O08553 Dihydropyrimidinase-related protein 2 Dpysl2 62 49.7 O08599 Syntaxin-binding protein 1 Stxbp1 68 35.4 O08663 Methionine aminopeptidase 2 Metap2 53 13.2 O08749 Dihydrolipoyl dehydrogenase, mitochondrial Dld 54 11 O08788 Dynactin subunit 1 Dctn1 142 17.7 O08848 60 kDa SS-A/Ro ribonucleoprotein Trove2 60 5.4 O09061 Proteasome subunit beta type-1 Psmb1 26 33.3 O35098 Dihydropyrimidinase-related protein 4 Dpysl4 62 15.7 O35136 Neural cell adhesion molecule 2 Ncam2 93 4.8 O35226 26S proteasome non-ATPase regulatory subunit 4 Psmd4 41 14.4 O35286 Pre-mRNA-splicing factor ATP-dependent RNA helicase Dhx15 91 2.9 DHX15 O35464 Semaphorin-6A Sema6a 114 4.8 O35593 26S proteasome non-ATPase regulatory subunit 14 Psmd14 35 9.4 O35685 Nuclear migration protein nudC Nudc 38 19.6 O35841 Apoptosis inhibitor 5 Api5 57 16.3 O35864 COP9 signalosome complex subunit 5 Cops5 38 17.7 O54829 Regulator of G-protein signaling 7 Rgs7 55 5.1 O55013 Trafficking protein particle complex subunit 3 Trappc3 20 10 O55100 Synaptogyrin-1 Syngri1 26 10.3 O55131 Septin-7 Sept7 51 17.2 O55234 Proteasome subunit beta type-5 Psmb5 29 20.8 O70194 Eukaryotic translation initiation factor 3 subunit D Eif3d 64 4 O70310 Glycylpeptide N-tetradecanoyltransferase 1 Nmt1 57 22.8 O70311 Glycylpeptide N-tetradecanoyltransferase 2 Nmt2 60 8.3 O70435 Proteasome subunit alpha type-3 Psma3 28 30.6 O70493 Sorting nexin-12 Snx12 19 12.7 O88342 WD repeat-containing protein 1 Wdr1 66 30.5 O88447 Kinesin light chain 1 Klcl 61 6.7 O88485 Cytoplasmic dynein 1 intermediate chain 1 Dync1i1 71 12.7 O88487 Cytoplasmic dynein 1 intermediate chain 2 Dync1i2 68 7.2 O88543 COP9 signalosome complex subunit 3 Cops3 48 9 O88544 COP9 signalosome complex subunit 4 Cops4 46 14 O88569 Heterogeneous nuclear ribonucleoproteins A2/B1 Hnrnpa2b1 37 21.8 O88643 Serine/threonine-protein kinase PAK 1 Pak1 61 20.9 O88685 26S protease regulatory subunit 6A Psmc3 50 27.6 O88735 Ensconsin Map7 82 5.2 O88844 Isocitrate dehydrogenase [NADP] cytoplasmic Idh1 47 23.9 O88935 Synapsin-1 Syn1 74 10.3 P00920 Carbonic anhydrase 2 Ca2 29 5.4 P01027 Complement C3 C3 186 2.9 P01869 Ig gamma-1 chain C region, membrane-bound form Ighg1 43 5.6 P02088 Hemoglobin subunit beta-1 Hbb-b1 16 12.2 P03995 Glial fibrillary acidic protein Gfap 50 15.8 P04370 Myelin basic protein Mbp 27 8 P05063 Fructose-bisphosphate aldolase C Aldoc 39 41 P05064 Fructose-bisphosphate aldolase A Aldoa 39 35.4 P05132 cAMP-dependent protein kinase catalytic subunit alpha Prkaca 41 29.1 P05202 Aspartate aminotransferase, mitochondrial Got2 47 15.1 P06151 F-lactate dehydrogenase A chain Ldha 36 8.7 P06745 Glucose-6-phosphate isomerase Gpi 63 26 P07356 Annexin A2 Anxa2 39 34.5 P07901 Heat shock protein HSP 90-alpha Hsp90aa1 85 15.3 P08113 Endoplasmic Hsp90b1 92 23.2 P08249 Malate dehydrogenase, mitochondrial Mdh2 36 26.9 P08551 Neurofilament light polypeptide Neff 62 35.5 P08553 Neurofilament medium polypeptide Nefm 96 20.5 P09041 Phosphoglycerate kinase 2 Pgk2 45 25.9 P09405 Nucleolin Ncl 77 3.3 P09411 Phosphoglycerate kinase 1 Pgk1 45 47.2 P0CG49 Polyubiquitin-B Ubb 9 44.2 P10107 Annexin A1 Anxa1 39 7.8 P10126 Elongation factor 1-alpha 1 Eef1a1 50 37.7 P10630 Eukaryotic initiation factor 4A-II Eif4a2 46 33.4 P10637 Microtubule-associated protein tau Mapt 76 17.6 P10711 Transcription elongation factor A protein 1 Tcea1 34 26.9 P11103 Poly[ADP-ribose] polymerase 1 Parp1 113 7.6 P11247 Myeloperoxidase Mpo 81 3.3 P11499 Heat shock

protein HSP 90ab1 83 17.1 P11798 Calcium/calmodulin-dependent protein kinase type II
 Camk2a 54 6.9 subunit alpha P11983 T-complex protein 1 subunit alpha Tcp1 60 10.4 P12367 cAMP-
 dependent protein kinase type II-alpha Prkar2a 45 12.2 regulatory subunit P12382 ATP-dependent 6-
 phosphofructokinase, liver type Pfkf 85 15.5 P12960 Contactin-1 Cntn1 113 11.9 P14152 Malate
 dehydrogenase, cytoplasmic Mdh1 37 9 P14211 Calreticulin Calr 48 11.5 P14685 26S proteasome non-
 ATPase regulatory subunit 3 Psmd3 61 16.8 P14824 Annexin A6 Anxa6 76 20.5 P14873 Microtubule-
 associated protein 1B Map1b 270 11.5 P15105 Glutamine synthetase Glul 42 8.3 P16330 2',3'-cyclic-
 nucleotide 3'-phosphodiesterase Cnp 47 8.1 P16546 Spectrin alpha chain, non-erythrocytic 1 Sptan1 285
 6 P16627 Heat shock 70 kDa protein 1-like Hspa1l 71 23.4 P16858 Glyceraldehyde-3-phosphate
 dehydrogenase Gapdh 36 26.4 P17095 High mobility group protein HMG-I/HMG-Y Hmga1 12 26.2
 P17156 Heat shock-related 70 kDa protein 2 Hspa2 70 45 P17182 Alpha-enolase Enol 47 35.5 P17183
 Gamma-enolase Eno2 47 11.8 P17426 AP-2 complex subunit alpha-1 Ap2a1 108 11.4 P17427 AP-2
 complex subunit alpha-2 Ap2a2 104 3 P17742 Peptidyl-prolyl cis-trans isomerase A Ppia 18 23.2 P17751
 Triosephosphate isomerase Tpi1 32 14 P18760 Cofilin-1 Cfl1 19 25.9 P19246 Neurofilament heavy
 polypeptide Nefh 117 5.8 P20029 78 kDa glucose-regulated protein Hspa5 72 40.3 P20357 Microtubule-
 associated protein 2 Map2 199 15.9 P21550 Beta-enolase Eno3 47 24.2 P24369 Peptidyl-prolyl cis-trans
 isomerase B Ppib 24 30.1 P26040 Ezrin Ezr 69 15.5 P26041 Moesin Msn 68 39 P26043 Radixin Rdx 69
 38.6 P26443 Glutamate dehydrogenase 1, mitochondrial Glud1 61 10 P26516 26S proteasome non-
 ATPase regulatory subunit 7 Psmd7 37 13.1 P26638 Serine--tRNA ligase, cytoplasmic Sars 58 20.1
 P27546 Microtubule-associated protein 4 Map4 117 3.6 P27773 Protein disulfide-isomerase A3 Pdia3 57
 19.6 P28271 Cytoplasmic aconitate hydratase Aco1 98 2.9 P28352 DNA-(apurinic or apyrimidinic site)
 lyase Apex1 35 18.9 P28481 Collagen alpha-1(II) chain Col2a1 142 1.6 P28650 Adenylosuccinate
 synthetase isozyme 1 Adssl1 50 24.9 P28652 Calcium/calmodulin-dependent protein kinase type II
 Camk2b 60 3.9 subunit beta P28660 Nck-associated protein 1 Nckap1 129 11.8 P28663 Beta-soluble
 NSF attachment protein Napb 34 17.8 P28738 Kinesin heavy chain isoform 5C Kif5c 109 25 P28740
 Kinesin-like protein KIF2A Kif2a 80 19.7 P30416 Peptidyl-prolyl cis-trans isomerase FKBP4 Fkbp4 52 9
 P31230 Aminoacyl tRNA synthase complex-interacting Aimp1 34 9.7 multifunctional protein 1 P31324
 cAMP-dependent protein kinase type II-beta regulatory Prkar2b 46 29.1 subunit P31938 Dual specificity
 mitogen-activated protein kinase kinase Map2k1 43 24.2 1 P32883 GTPase KRas Kras 22 21.2 P32921
 Tryptophan--tRNA ligase, cytoplasmic Wars 54 11.2 P34152 Focal adhesion kinase 1 Ptk2 124 2 P35235
 Tyrosine-protein phosphatase non-receptor type 11 Ptpn11 68 25.8 P35700 Peroxiredoxin-1 Prdx1 22
 27.1 P36916 Guanine nucleotide-binding protein-like 1 Gnl1 69 8.4 P37804 Transgelin Tagln 23 17.4
 P38647 Stress-70 protein, mitochondrial Hspa9 73 12.8 P39053 Dynamin-1 Dnm1 98 19.7 P39054
 Dynamin-2 Dnm2 98 7 P39749 Flap endonuclease 1 Fen1 42 8.5 P40124 Adenylyl cyclase-associated
 protein 1 Cap1 52 16.7 P40142 Transketolase Tkt 68 17.3 P42669 Transcriptional activator protein Pur-
 alpha Pura 35 15.3 P42932 T-complex protein 1 subunit theta Cct8 60 13.1 P45591 Cofilin-2 Cfl2 19
 18.7 P45878 Peptidyl-prolyl cis-trans isomerase FKBP2 Fkbp2 15 17.1 P46096 Synaptotagmin-1 Syt1 47
 33 P46460 Vesicle-fusing ATPase Nsf 83 14 P46471 26S protease regulatory subunit 7 Psmc2 49 4.8
 P46660 Alpha-internexin Ina 55 23.8 P46664 Adenylosuccinate synthetase isozyme 2 Adss 50 6.4
 P47199 Quinone oxidoreductase Cryz 35 5.7 P47708 Rabphilin-3A Rph3a 75 4 P47753 F-actin-capping
 protein subunit alpha-1 Capza1 33 22.4 P47754 F-actin-capping protein subunit alpha-2 Capza2 33 51.7
 P47757 F-actin-capping protein subunit beta Capzb 31 20.9 P47857 ATP-dependent 6-
 phosphofructokinase, muscle type Pfkf 85 17.7 P47934 Carnitine O-acetyltransferase Crat 71 2.7
 P48024 Eukaryotic translation initiation factor 1 Eif1 13 55.8 P48036 Annexin A5 Anxa5 36 6.3 P48722
 Heat shock 70 kDa protein 4L Hspa4l 94 8.5 P48758 Carbonyl reductase [NADPH] 1 Cbr1 31 8.3
 P49182 Heparin cofactor 2 Serpind1 54 2.9 P49312 Heterogeneous nuclear ribonucleoprotein A1
 Hnrnpa1 34 17.2 P49615 Cyclin-dependent-like kinase 5 Cdk5 33 44.9 P49722 Proteasome subunit alpha
 type-2 Psma2 26 42.7 P50516 V-type proton ATPase catalytic subunit A Atp6v1a 68 29.5 P50518 V-type
 proton ATPase subunit E 1 Atp6v1e1 26 29.2 P50580 Proliferation-associated protein 2G4 Pa2g4 44 33.5
 P51174 Long-chain specific acyl-CoA dehydrogenase, Acadl 48 6 mitochondrial P51432 1-
 phosphatidylinositol 4,5-bisphosphate Plcb3 139 1.6 phosphodiesterase beta-3 P51859 Hepatoma-derived
 growth factor Hdgf 26 14.8 P52196 Thiosulfate sulfurtransferase Tst 33 8.8 P52480 Pyruvate kinase
 PKM Pkm 58 35.6 P54071 Isocitrate dehydrogenase [NADP], mitochondrial Idh2 51 22.6 P54227

Stathmin Stmn1 17 14.8 P54823 ATP-dependent RNA helicase DDX6 Ddx6 54 4.8 P55066
 Neurocan core protein Ncan 137 1.5 P55821 Stathmin-2 Stmn2 21 10.6 P56212 cAMP-regulated
 phosphoprotein 19 Arpp19 12 49.1 P56399 Ubiquitin carboxyl-terminal hydrolase 5 Usp5 96 2.4 P57759
 Endoplasmic reticulum resident protein 29 Erp29 29 6.9 P58252 Elongation factor 2 Eef2 95 22.7
 P59325 Eukaryotic translation initiation factor 5 Eif5 49 23.8 P60521 Gamma-aminobutyric acid
 receptor-associated Gabarapl2 14 36.8 protein-like 2 P60710 Actin, cytoplasmic 1 Actb 42 48 P60840
 Alpha-endosulfine Ensa 13 30.6 P60843 Eukaryotic initiation factor 4A-I Eif4a1 46 17.5 P61082
 NEDD8-conjugating enzyme Ubc12 Ube2m 21 16.4 P61089 Ubiquitin-conjugating enzyme E2 N Ube2n
 17 43.4 P61148 Fibroblast growth factor 1 Fgf1 17 20 P61161 Actin-related protein 2 Actr2 45 22.8
 P61164 Alpha-centractin Actr1a 43 28.7 P61202 COP9 signalosome complex subunit 2 Cops2 52 13.5
 P61222 ATP-binding cassette sub-family E member 1 Abce1 67 23.7 P61329 Fibroblast growth factor 12
 Fgf12 27 22.2 P61961 Ubiquitin-fold modifier 1 Ufm1 9 50.6 P61965 WD repeat-containing protein 5
 Wdr5 37 13.5 P61982 14-3-3 protein gamma Ywhag 28 33.2 P62082 40S ribosomal protein S7 Rps7 22
 32.8 P62137 Serine/threonine-protein phosphatase PP1-alpha Ppp1ca 38 13 catalytic subunit P62141
 Serine/threonine-protein phosphatase PP1-beta Ppp1cb 37 17.7 catalytic subunit P62196 26S protease
 regulatory subunit 8 Psmc5 46 9.6 P62204 Calmodulin Calm1 17 33.6 P62245 40S ribosomal protein
 S15a Rps15a 15 23.8 P62334 26S protease regulatory subunit 10B Psmc6 44 7.5 P62627 Dynein light
 chain roadblock-type 1 Dynlrb1 11 21.9 P62631 Elongation factor 1-alpha 2 Eef1a2 50 55.3 P62806
 Histone H4 Hist1h4a 11 17.5 P62814 V-type proton ATPase subunit B, brain isoform Atp6v1b2 57 5.1
 P62827 GTP-binding nuclear protein Ran Ran 24 19.9 P62852 40S ribosomal protein S25 Rps25 14 14.4
 P62858 40S ribosomal protein S28 Rps28 8 30.4 P62881 Guanine nucleotide-binding protein subunit
 beta-5 Gnb5 44 8.4 P63001 Ras-related C3 botulinum toxin substrate 1 Rac1 21 16.7 P63005 Platelet-
 activating factor acetylhydrolase IB subunit Pafah1b1 47 39.8 alpha P63017 Heat shock cognate 71 kDa
 protein Hspa8 71 56.7 P63028 Translationally-controlled tumor protein Tpt1 19 18.6 P63046
 Sulfotransferase 4A1 Sult4a1 33 14.1 P63085 Mitogen-activated protein kinase 1 Mapk1 41 37.4 P63087
 Serine/threonine-protein phosphatase PP1-gamma Ppp1cc 37 12.1 catalytic subunit P63101 14-3-3
 protein zeta/delta Ywhaz 28 22.4 P63158 High mobility group protein B1 Hmgb1 25 21.9 P63242
 Eukaryotic translation initiation factor 5A-1 Eif5a 17 34.4 P63280 SUMO-conjugating enzyme UBC9
 Ube2i 18 16.5 P63328 Serine/threonine-protein phosphatase 2B catalytic Ppp3ca 59 3.5 subunit alpha
 isoform P68033 Actin, alpha cardiac muscle 1 Actc1 42 27.3 P68037 Ubiquitin-conjugating enzyme E2
 L3 Ube2l3 18 51.9 P68181 cAMP-dependent protein kinase catalytic subunit beta Prkacb 41 22.8 P68254
 14-3-3 protein theta Ywhaq 28 11.8 P68368 Tubulin alpha-4A chain Tuba4a 50 33.3 P68372 Tubulin
 beta-4B chain Tubb4b 50 48.8 P68373 Tubulin alpha-1C chain Tuba1c 50 43 P70122 Ribosome
 maturation protein SBDS Sbd5 29 40.4 P70236 Dual specificity mitogen-activated protein kinase kinase
 Map2k6 37 20.1 P70296 Phosphatidylethanolamine-binding protein 1 Pebp1 21 17.6 P70336 Rho-
 associated protein kinase 2 Rock2 161 9.1 P70441 Na(+)/H(+) exchange regulatory cofactor NHE-RF1
 Slc9a3r1 39 7 P70670 Nascent polypeptide-associated complex subunit Naca 221 2.6 alpha, muscle-
 specific form P70695 Fructose-1,6-bisphosphatase isozyme 2 Fbp2 37 6.8 P80313 T-complex protein 1
 subunit eta Cct7 60 16.4 P80314 T-complex protein 1 subunit beta Cct2 57 25 P80315 T-complex protein
 1 subunit delta Cct4 58 7.4 P80316 T-complex protein 1 subunit epsilon Cct5 60 9.6 P80317 T-complex
 protein 1 subunit zeta Cct6a 58 18.6 P80318 T-complex protein 1 subunit gamma Cct3 61 25.5 P84078
 ADP-ribosylation factor 1 Arf1 21 43.6 P84091 AP-2 complex subunit mu Ap2m1 50 4.8 P97376 Protein
 FRG1 Frg1 29 15.5 P97390 Vacuolar protein sorting-associated protein 45 Vps45 65 7.2 P97427
 Dihydropyrimidinase-related protein 1 Crmp1 62 21.7 P97807 Fumarate hydratase, mitochondrial Fh 54
 13 P99024 Tubulin beta-5 chain Tubb5 50 48.9 P99026 Proteasome subunit beta type-4 Psmb4 29 8.7
 Q00PI9 Heterogeneous nuclear ribonucleoprotein U-like Hnrnpul2 85 2.6 protein 2 Q01730 Ras
 suppressor protein 1 Rsu1 32 27.8 Q02053 Ubiquitin-like modifier-activating enzyme 1 Uba1 118 4
 Q04447 Creatine kinase B-type Ckb 43 13.6 Q05BC3 Echinoderm microtubule-associated protein-like 1
 Eml1 90 18.7 Q06138 Calcium-binding protein 39 Cab39 40 24.9 Q08642 Protein-arginine deiminase
 type-2 Padi2 76 8.3 Q2NL51 Glycogen synthase kinase-3 alpha Gsk3a 52 7.1 Q2PFD7 PH and SEC7
 domain-containing protein 3 Psd3 115 5.2 Q3TGF2 Protein FAM107B Fam107b 16 16 Q3THG9 Alanyl-
 tRNA editing protein Aarsd1 Aarsd1 61 4.4 Q3THK3 General transcription factor IIF subunit 1 Gtf2f1 57
 6.7 Q3THK7 GMP synthase [glutamine-hydrolyzing] Gmps 77 6.2 Q3TKT4 Transcription activator

BRG1 181 1.4 Q3TXS7 26S proteasome non-ATPase regulatory subunit 1 Psmd1 106 9.4
 Q3UGR5 Haloacid dehalogenase-like hydrolase domain- Hdh2 29 11.2 containing protein 2 Q3UHF1
 CaM kinase-like vesicle-associated protein Camkv 55 20.1 Q3UHX2 28 kDa heat- and acid-stable
 phosphoprotein Pdap1 21 27.1 Q3ULJ0 Glycerol-3-phosphate dehydrogenase 1-like protein Gpd1l 38 4.6
 Q3UM45 Protein phosphatase 1 regulatory subunit 7 Ppp1r7 41 10.2 Q3UMU9 Hepatoma-derived
 growth factor-related protein 2 Hdgrp2 74 9.3 Q3UMY5 Echinoderm microtubule-associated protein-
 like 4 Eml4 110 2.8 Q3UV17 Keratin, type II cytoskeletal 2 oral Krt76 63 2.7 Q3UX10 Tubulin alpha
 chain-like 3 Tubal3 50 6.3 Q3V1L4 Cytosolic purine 5'-nucleotidase Nt5c2 65 4.8 Q4KMM3 Oxidation
 resistance protein 1 Oxr1 96 4.7 Q5M8N0 CB1 cannabinoid receptor-interacting protein 1 Cnrip1 19 5.5
 Q5SQX6 Cytoplasmic FMR1-interacting protein 2 Cyfip2 146 12.8 Q5SSL4 Active breakpoint cluster
 region-related protein Abr 98 2.2 Q60668 Heterogeneous nuclear ribonucleoprotein D0 Hnrnpd 38 32.1
 Q60676 Serine/threonine-protein phosphatase 5 Ppp5c 57 24.2 Q60692 Proteasome subunit beta type-6
 Psmb6 25 8 Q60864 Stress-induced-phosphoprotein 1 Stip1 63 23.4 Q60872 Eukaryotic translation
 initiation factor 1A Eif1a 16 18.1 Q60875 Rho guanine nucleotide exchange factor 2 Arhgef2 112 4.3
 Q60900 ELAV-like protein 3 Elavl3 40 4.4 Q60972 Histone-binding protein RBBP4 Rbbp4 48 16.2
 Q61035 Histidine--tRNA ligase, cytoplasmic Hars 57 14.9 Q61036 Serine/threonine-protein kinase PAK
 3 Pak3 62 11.1 Q61142 Spindlin-1 Spin1 30 10.3 Q61166 Microtubule-associated protein RP/EB family
 member Mapre1 30 21.3 1 Q61316 Heat shock 70 kDa protein 4 Hspa4 94 2.9 Q61425 Hydroxyacyl-
 coenzyme A dehydrogenase, Hadh 34 6.7 mitochondrial Q61548 Clathrin coat assembly protein AP180
 Snap91 92 12.7 Q61553 Fascin Fscn1 55 19.3 Q61644 Protein kinase C and casein kinase substrate in
 neurons Pacsin1 51 12.5 protein 1 Q61646 Haptoglobin Hp 39 4.9 Q61696 Heat shock 70 kDa protein 1A
 Hspa1a 70 32.3 Q61753 D-3-phosphoglycerate dehydrogenase Phgdh 57 7.3 Q61768 Kinesin-1 heavy
 chain Kif5b 110 15.9 Q61879 Myosin-10 Myh10 229 3.8 Q62165 Dystroglycan Dag1 97 3.1 Q62188
 Dihydropyrimidinase-related protein 3 Dpysl3 62 24.7 Q62261 Spectrin beta chain, non-erythrocytic 1
 Sptbn1 274 1.2 Q62420 Endophilin-A1 Sh3gl2 40 15.3 Q62446 Peptidyl-prolyl cis-trans isomerase
 FKBP3 Fkbp3 25 25.9 Q63844 Mitogen-activated protein kinase 3 Mapk3 43 15.5 Q63912
 Oligodendrocyte-myelin glycoprotein Omg 49 9.5 Q64152 Transcription factor BTF3 Btf3 22 32.4
 Q641P0 Actin-related protein 3B Actr3b 48 12.9 Q64467 Glyceraldehyde-3-phosphate dehydrogenase,
 testis- Gapdhs 48 4.1 specific Q64514 Tripeptidyl-peptidase 2 Tpp2 140 3.7 Q64669 NAD(P)H
 dehydrogenase [quinone] 1 Nqo1 31 6.9 Q68FL6 Methionine--tRNA ligase, cytoplasmic Mars 101 3.2
 Q69ZS7 HBS1-like protein Hbs1l 75 8.4 Q6A028 Switch-associated protein 70 Swap70 69 12.6 Q6DIC0
 Probable global transcription activator SNF2L2 Smarca2 180 1.6 Q6NZZB0 DnaJ homolog subfamily C
 member 8 Dnajc8 30 36 Q6P1B1 Xaa-Pro aminopeptidase 1 Xpnpep1 70 14.6 Q6PDI5 Proteasome-
 associated protein ECM29 homolog Ecm29 204 5.9 Q6PDL0 Cytoplasmic dynein 1 light intermediate
 chain 2 Dync1li2 54 8.9 Q6PER3 Microtubule-associated protein RP/EB family member Mapre3 32 35.6
 3 Q6PGN3 Serine/threonine-protein kinase DCLK2 Dclk2 83 6.1 Q6WVG3 BTB/POZ domain-
 containing protein KCTD12 Kctd12 36 7.3 Q6ZPJ3 (E3-independent) E2 ubiquitin-conjugating enzyme
 Ube2o 141 1.8 UBE2O Q6ZQ38 Cullin-associated NEDD8-dissociated protein 1 Cand1 136 12.1
 Q6ZWX6 Eukaryotic translation initiation factor 2 subunit 1 Eif2s1 36 44.8 Q71LX4 Talin-2 Tln2 254 1
 Q78JW9 Ubiquitin domain-containing protein UBFD1 Ubfd1 40 11.4 Q78PG9 Coiled-coil domain-
 containing protein 25 Ccdc25 24 17.3 Q78ZA7 Nucleosome assembly protein 1-like 4 Nap1l4 43 7.5
 Q792Z1 MCG140784 Try10 26 8.1 Q7M6Y3 Phosphatidylinositol-binding clathrin assembly protein
 Picalm 72 6.7 Q7TMB8 Cytoplasmic FMR1-interacting protein 1 Cyfip1 145 7.7 Q7TMK9
 Heterogeneous nuclear ribonucleoprotein Q Syncrip 70 8.2 Q7TMM9 Tubulin beta-2A chain Tubb2a 50
 50.8 Q7TNG5 Echinoderm microtubule-associated protein-like 2 Eml2 71 4.5 Q7TNV0 Protein DEK
 Dek 43 6.6 Q7TQD2 Tubulin polymerization-promoting protein Tppp 24 16.1 Q7TSJ2 Microtubule-
 associated protein 6 Map6 96 3.9 Q80TV8 CLIP-associating protein 1 Clasp1 169 10.2 Q80UG5 Septin-
 9 Sept9 66 4.3 Q80UM3 N-alpha-acetyltransferase 15, NatA auxiliary subunit Naa15 101 9.9 Q80VP1
 Epsin-1 Epn1 60 3.8 Q80XU3 Nuclear ubiquitous casein and cyclin-dependent kinase Nucks1 26 19.7
 substrate 1 Q810S1 Calcium uniporter regulatory subunit MCUB, Mcub 40 5.5 mitochondrial Q810U3
 Neurofascin Nfasc 138 6 Q8BFR5 Elongation factor Tu, mitochondrial Tufm 50 5.8 Q8BFZ3 Beta-actin-
 like protein 2 Actbl2 42 22.1 Q8BG05 Heterogeneous nuclear ribonucleoprotein A3 Hnrnpa3 40 24.8
 Q8BG32 26S proteasome non-ATPase regulatory subunit 11 Psmd11 47 32 Q8BGA3 Leucine-rich repeat

membrane-membrane protein 2 Lrrtm2 59 5.4 Q8BGQ7 tRNA ligase, cytoplasmic Aars 107
 2.4 Q8BGR9 Ubiquitin-like domain-containing CTD phosphatase 1 Ublcp1 37 20.1 Q8BGT8 Phytanoyl-CoA hydroxylase-interacting protein-like Phyhipl 42 14.7 Q8BGY2 Eukaryotic translation initiation factor 5A-2 Eif5a2 17 19 Q8BH57 WD repeat-containing protein 48 Wdr48 76 11.2 Q8BJ37 Tyrosyl-DNA phosphodiesterase 1 Tdp1 69 5.1 Q8BJD1 Inter-alpha-trypsin inhibitor heavy chain H5 Itih5 107 3.2 Q8BK63 Casein kinase I isoform alpha Csnk1a1 39 10.4 Q8BK64 Activator of 90 kDa heat shock protein ATPase Ahsa1 38 18.9 homolog 1 Q8BK67 Protein RCC2 Rcc2 56 14.8 Q8BKG3 Inactive tyrosine-protein kinase 7 Ptk7 118 2.8 Q8BKX1 Brain-specific angiogenesis inhibitor 1-associated Baiap2 59 6.7 protein 2 Q8BLJ3 PI-PLC X domain-containing protein 3 Plcxd3 36 10.3 Q8BMF3 NADP-dependent malic enzyme, mitochondrial Me3 67 3.1 Q8BMJ2 Leucine--tRNA ligase, cytoplasmic Lars 134 2.6 Q8BP47 Asparagine--tRNA ligase, cytoplasmic Nars 64 5.5 Q8BRT1 CLIP-associating protein 2 Clasp2 141 9.3 Q8BU30 Isoleucine--tRNA ligase, cytoplasmic Iars 144 2.9 Q8BVI4 Dihydropteridine reductase Qdpr 26 15.8 Q8BVQ5 Protein phosphatase methylesterase 1 Ppme1 42 8.8 Q8BVU5 ADP-ribose pyrophosphatase, mitochondrial Nudt9 39 6.6 Q8BW96 Calcium/calmodulin-dependent protein kinase type 1D Camk1d 43 16.1 Q8BWG8 Beta-arrestin-1 Arrb1 47 10.8 Q8BWR2 PITH domain-containing protein 1 Pithd1 24 28.4 Q8BWY3 Eukaryotic peptide chain release factor subunit 1 Etf1 49 10.1 Q8BWZ3 N-alpha-acetyltransferase 25, NatB auxiliary subunit Naa25 112 6.8 Q8BYB9 Protein O-glucosyltransferase 1 Poglut1 46 11.7 Q8BZ98 Dynamin-3 Dnm3 97 8.8 Q8C1B1 Calmodulin-regulated spectrin-associated protein 2 Camsap2 164 1.6 Q8C1B7 Septin-11 Sept11 50 18.8 Q8C1W1 Vasohibin-1 Vash1 42 12.8 Q8C4Q6 Axin interactor, dorsalization-associated protein Aida 35 16.4 Q8C5R8 Phosphoribosyl pyrophosphate synthetase 1-like 1 Prps1l1 35 11.6 Q8C8R3 Ankyrin-2 Ank2 426 1.2 Q8CBY8 Dynactin subunit 4 Dctn4 53 7.3 Q8CDN6 Thioredoxin-like protein 1 Txnl1 32 30.1 Q8CGC7 Bifunctional glutamate/proline--tRNA ligase Eprs 170 2 Q8CGF7 Transcription elongation regulator 1 Tcerg1 124 3.9 Q8CHC4 Synaptojanin-1 Synj1 173 13 Q8CIB5 Fermitin family homolog 2 Fermt2 78 3.2 Q8CIN4 Serine/threonine-protein kinase PAK 2 Pak2 58 11.5 Q8JZK9 Hydroxymethylglutaryl-CoA synthase, cytoplasmic Hmgcs1 58 6.9 Q8JZQ9 Eukaryotic translation initiation factor 3 subunit B Eif3b 91 3.9 Q8K0S0 Phytanoyl-CoA hydroxylase-interacting protein Phyhip 38 8.2 Q8K0U4 Heat shock 70 kDa protein 12A Hspa12a 75 9.6 Q8K1J6 CCA tRNA nucleotidyltransferase 1, mitochondrial Trnt1 50 9.2 Q8K1M6 Dynamin-1-like protein Dnm1l 83 11.7 Q8K2T1 NmrA-like family domain-containing protein 1 Nmral1 34 8.1 Q8K409 DNA polymerase beta Polb 38 13.7 Q8QZT1 Acetyl-CoA acetyltransferase, mitochondrial Acat1 45 9.9 Q8QZY1 Eukaryotic translation initiation factor 3 subunit L Eif3l 67 8.2 Q8R001 Microtubule-associated protein RP/EB family member Mapre2 37 21.2 2 Q8R050 Eukaryotic peptide chain release factor GTP-binding Gspt1 69 9 subunit ERF3A Q8R0F6 Integrin-linked kinase-associated serine/threonine Ilkap 43 18.1 phosphatase 2C Q8R0Y6 Cytosolic 10-formyltetrahydrofolate dehydrogenase Aldh1l1 99 3.1 Q8R1B4 Eukaryotic translation initiation factor 3 subunit C Eif3c 106 7.7 Q8R1Q8 Cytoplasmic dynein 1 light intermediate chain 1 Dync1li1 57 5.4 Q8R3R8 Gamma-aminobutyric acid receptor-associated Gabarapl1 14 35 protein-like 1 Q8R574 Phosphoribosyl pyrophosphate synthase-associated Prpsap2 41 28.2 protein 2 Q8R5C5 Beta-actin Actr1b 42 20.5 Q8R5H6 Wiskott-Aldrich syndrome protein family member 1 Wasf1 62 3.6 Q8VDD5 Myosin-9 Myh9 226 5 Q8VDM4 26S proteasome non-ATPase regulatory subunit 2 Psmd2 100 13.4 Q8VE37 Regulator of chromosome condensation Rcc1 45 18.3 Q8VED9 Galectin-related protein Lgalsl 19 25.6 Q8VEK3 Heterogeneous nuclear ribonucleoprotein U Hnrnpu 88 11.4 Q8VHM5 Heterogeneous nuclear ribonucleoprotein R Hnrnpr 71 6 Q91UZ1 Phosphoinositide phospholipase C Plcb4 135 6.7 Q91V09 WD repeat-containing protein 13 Wdr13 54 7.8 Q91V12 Cytosolic acyl coenzyme A thioester hydrolase Acot7 43 18.6 Q91V57 N-chimaerin Chn1 53 6.1 Q91V89 Protein phosphatase 2, regulatory subunit B (B56), Ppp2r5d 69 4.2 delta isoform Q91V92 ATP-citrate synthase Acly 120 15.4 Q91VK1 Basic leucine zipper and W2 domain-containing protein Bzw2 48 19.8 2 Q91VR5 ATP-dependent RNA helicase DDX1 Ddx1 82 8.4 Q91VR7 Microtubule-associated proteins 1A/1B light chain 3A Map1lc3a 14 23.1 Q91VZ6 Stromal membrane-associated protein 1 Smap1 48 4.5 Q91WC0 Histone-lysine N-methyltransferase setd3 Setd3 67 3 Q91WQ3 Tyrosine--tRNA ligase, cytoplasmic Yars 59 37.7 Q91XL9 Oxysterol-binding protein-related protein 1 Osbpl1a 108 7.2 Q91XM9 Disks large homolog 2 Dlg2 95 16.2 Q91YE3 Egl nine homolog 1 Egl1 43 5.2 Q91YJ3 Thymocyte nuclear protein 1 Thyn1 26 7.5 Q91YP2 Neurolysin, mitochondrial Nln 80 6.4 Q91YR1 Twinfilin-1 Twf1

40 15.4 Q91ZJ5 Uridylyltransferase Ugp2 57 22 Q91ZW3 SWI/SNF-related matrix-associated actin-dependent Smarca5 122 2 regulator of chromatin subfamily A member 5
 Q921M7 Protein FAM49B Fam49b 37 38 Q921W0 Charged multivesicular body protein 1a Chmp1a 22 12.8 Q921X9 Protein disulfide-isomerase A5 Pdia5 59 6.6 Q922B2 Aspartate--tRNA ligase, cytoplasmic Dars 57 5.6 Q922D8 C-1-tetrahydrofolate synthase, cytoplasmic Mthfd1 101 7.1 Q922F4 Tubulin beta-6 chain Tubb6 50 26.8 Q922J3 CAP-Gly domain-containing linker protein 1 Clip1 156 12.9 Q923D2 Flavin reductase (NADPH) Blvrb 22 21.4 Q924Y0 Gamma-butyrobetaine dioxygenase Bbox1 45 6.5 Q99020 Heterogeneous nuclear ribonucleoprotein A/B Hnrnpab 31 20 Q99104 Unconventional myosin-Va Myo5a 216 1.1 Q99J08 SEC14-like protein 2 Sec14l2 46 5.2 Q99J36 THUMP domain-containing protein 1 Thumpd1 39 19.1 Q99J77 Sialic acid synthase Nans 40 8.9 Q99JF8 PC4 and SFRS1-interacting protein Psip1 60 14.8 Q99JI4 26S proteasome non-ATPase regulatory subunit 6 Psmd6 46 11.8 Q99JY9 Actin-related protein 3 Actr3 47 30.1 Q99K85 Phosphoserine aminotransferase Psat1 40 7.8 Q99KB8 Hydroxyacylglutathione hydrolase, mitochondrial Hagh 34 17.2 Q99KI0 Aconitate hydratase, mitochondrial Aco2 85 27.2 Q99KJ8 Dynactin subunit 2 Dctn2 44 17.7 Q99KK2 N-acylneuraminate cytidylyltransferase Cmas 48 4.6 Q99L45 Eukaryotic translation initiation factor 2 subunit 2 Eif2s2 38 24.2 Q99LC8 Translation initiation factor eIF-2B subunit alpha Eif2b1 34 9.2 Q99LD4 COP9 signalosome complex subunit 1 Gps1 53 5.5 Q99LF4 tRNA-splicing ligase RtcB homolog Rtcb 55 24.2 Q99LU0 Charged multivesicular body protein 1b-1 Chmp1b1 22 6 Q99NF3 Centrosomal protein of 41 kDa Cep41 41 7 Q99PT1 Rho GDP-dissociation inhibitor 1 Arhgdia 23 38.2 Q9CQ65 S-methyl-5'-thioadenosine phosphorylase Mtap 31 10.2 Q9CQC6 Basic leucine zipper and W2 domain-containing protein Bzw1 48 27 1 Q9CQH7 Transcription factor BTF3 homolog 4 Btf3l4 17 55.7 Q9CQJ6 Density-regulated protein Denr 22 20.7 Q9CQV6 Microtubule-associated proteins 1A/1B light chain 3B Map1lc3b 15 29.6 Q9CQV8 14-3-3 protein beta/alpha Ywhab 28 21.1 Q9CR16 Peptidyl-prolyl cis-trans isomerase D Ppid 41 26.5 Q9CR29 Coiled-coil domain-containing protein 43 Ccdc43 25 12.2 Q9CRB6 Tubulin polymerization-promoting protein family Tppp3 19 29 member 3 Q9CRC8 Leucine-rich repeat-containing protein 40 Lrrc40 68 2.8 Q9CRD2 ER membrane protein complex subunit 2 Emc2 35 10.1 Q9CS42 Ribose-phosphate pyrophosphokinase 2 Prps2 35 23.3 Q9CVB6 Actin-related protein 2/3 complex subunit 2 Arpc2 34 29.7 Q9CWJ9 Bifunctional purine biosynthesis protein PURH Atic 64 11.3 Q9CX34 Protein SGT1 homolog Sugt1 38 18.8 Q9CXU9 Eukaryotic translation initiation factor 1b Eif1b 13 44.2 Q9CXW3 Calcyclin-binding protein Cacybp 27 21.8 Q9CXW4 60S ribosomal protein L11 Rpl11 20 12.6 Q9CY64 Biliverdin reductase A Blvra 34 7.1 Q9CYR6 Phosphoacetylglucosamine mutase Pgm3 59 4.6 Q9CYT6 Adenylyl cyclase-associated protein 2 Cap2 53 4.8 Q9CZ30 Obg-like ATPase 1 Ola1 45 39.6 Q9CZ44 NSFL1 cofactor p47 Nsfl1c 41 7 Q9CZD3 Glycine--tRNA ligase Gars 82 9.6 Q9CZT6 Protein CMSS1 Cms1 32 7.6 Q9CZU6 Citrate synthase, mitochondrial Cs 52 7.8 Q9CZW5 Mitochondrial import receptor subunit TOM70 Tomm70 68 8.3 Q9CZX8 40S ribosomal protein S19 Rps19 16 23.4 Q9D051 Pyruvate dehydrogenase E1 component subunit beta, Pdhb 39 12.3 mitochondrial Q9D0I9 Arginine--tRNA ligase, cytoplasmic Rars 76 5 Q9D0K2 Succinyl-CoA: 3-ketoacid coenzyme A transferase 1, Oxt1 56 7.9 mitochondrial Q9D0L8 mRNA cap guanine-N7 methyltransferase Rnmt 53 5.6 Q9D0M1 Phosphoribosyl pyrophosphate synthase-associated Prpsap1 39 22.8 protein 1 Q9D0R2 Threonine--tRNA ligase, cytoplasmic Tars 83 12.6 Q9D1J3 SAP domain-containing ribonucleoprotein Sarnp 24 20.5 Q9D1P4 Cysteine and histidine-rich domain-containing protein Chordc1 37 18.7 1 Q9D2M8 Ubiquitin-conjugating enzyme E2 variant 2 Ube2v2 16 60 Q9D2R0 Acetoacetyl-CoA synthetase Aacs 75 11.8 Q9D358 Low molecular weight phosphotyrosine protein Acp1 18 12.7 phosphatase Q9D6F9 Tubulin beta-4A chain Tubb4a 50 48.9 Q9D708 Putative uncharacterized protein S100a16 14 10.5 Q9D7G0 Ribose-phosphate pyrophosphokinase 1 Prps1 35 15.1 Q9D7H3 RNA 3'-terminal phosphate cyclase RtcA 39 11.2 Q9D8B3 Charged multivesicular body protein 4b Chmp4b 25 13.4 Q9D8N0 Elongation factor 1-gamma Eef1g 50 22.2 Q9D8W5 26S proteasome non-ATPase regulatory subunit 12 Psmd12 53 16.9 Q9D8Y0 EF-hand domain-containing protein D2 Efhd2 27 15.8 Q9DB16 Calcium-binding protein 39-like Cab39l 39 13.1 Q9DB27 Malignant T-cell-amplified sequence 1 Mcts1 21 21 Q9DBG3 AP-2 complex subunit beta Ap2b1 105 9.2 Q9DBP5 UMP-CMP kinase Cmpk1 22 23.5 Q9DCD0 6-phosphogluconate dehydrogenase, decarboxylating Pgd 53 18 Q9DCD6 Gamma-aminobutyric acid receptor-associated protein Gabarap 14 34.2 Q9DCL9 Multifunctional protein ADE2 Paics 47 5.9 Q9DCN2 NADH-cytochrome b5 reductase 3 Cyb5r3 34 8 Q9DD18 D-tyrosyl-tRNA(Tyr)

deacylase 1 Dtd1 23 18.2 Q9EQF6 Dihydropyrimidinase-related protein 5 Dpysl5 62 15.6 Q9EQX4
 Allograft inflammatory factor 1-like Aif1l 17 21.3 Q9ERD7 Tubulin beta-3 chain Tubb3 50 52.2
 Q9ERE7 LDLR chaperone MESD Mesdc2 25 26.3 Q9ERQ8 Carbonic anhydrase 7 Ca7 30 12.9
 Q9ESN6 Tripartite motif-containing protein 2 Trim2 81 26.6 Q9JHQ5 Leucine zipper transcription
 factor-like protein 1 Lztlf1 35 7.4 Q9JHU4 Cytoplasmic dynein 1 heavy chain 1 Dync1h1 532 3.7
 Q9JIF0 Protein arginine N-methyltransferase 1 Prmt1 42 6.2 Q9JJK2 LanC-like protein 2 Lancl2 51 8.4
 Q9JJZ2 Tubulin alpha-8 chain Tuba8 50 22.3 Q9JKK7 Tropomodulin-2 Tmod2 40 20.8 Q9JLM8
 Serine/threonine-protein kinase DCLK1 Dcl1 84 12.3 Q9JLV5 Cullin-3 Cul3 89 17.8 Q9JM76 Actin-
 related protein 2/3 complex subunit 3 Arpc3 21 24.7 Q9JMA1 Ubiquitin carboxyl-terminal hydrolase 14
 Usp14 56 17.6 Q9JMG1 Endothelial differentiation-related factor 1 Edf1 16 31.8 Q9JMG7 Hepatoma-
 derived growth factor-related protein 3 Hdgrfp3 22 32.2 Q9QUM9 Proteasome subunit alpha type-6
 Psma6 27 31.3 Q9QUP5 Hyaluronan and proteoglycan link protein 1 Hapln1 40 7.6 Q9QUR7 Peptidyl-
 prolyl cis-trans isomerase NIMA-interacting 1 Pin1 18 37 Q9QX11 Cytohesin-1 Cyth1 46 5 Q9QXL2
 Kinesin-like protein KIF21A Kif21a 187 4.6 Q9QY36 N-alpha-acetyltransferase 10 Naa10 27 9.4
 Q9QY76 Vesicle-associated membrane protein-associated Vapb 27 16 protein B Q9QYB8 Beta-adducin
 Add2 81 11.7 Q9QYC0 Alpha-adducin Add1 81 6.9 Q9QYR6 Microtubule-associated protein 1A Map1a
 300 10.7 Q9QZ73 DCN1-like protein 1 Dcun1d1 30 10.8 Q9QZD9 Eukaryotic translation initiation
 factor 3 subunit I Eif3i 36 12.6 Q9R0P4 Small acidic protein Smap 20 21.5 Q9R0Q6 Actin-related
 protein 2/3 complex subunit 1A Arpc1a 42 17.6 Q9R0Y5 Adenylate kinase isoenzyme 1 Ak1 22 12.9
 Q9R1P1 Proteasome subunit beta type-3 Psmb3 23 34.1 Q9R1P3 Proteasome subunit beta type-2 Psmb2
 23 12.9 Q9R1P4 Proteasome subunit alpha type-1 Psma1 30 21.7 Q9R1Q8 Transgelin-3 Tagln3 22 38.2
 Q9R1R2 Tripartite motif-containing protein 3 Trim3 81 14.9 Q9WTN0 Geranylgeranyl pyrophosphate
 synthase Ggps1 35 9.7 Q9WTX5 S-phase kinase-associated protein 1 Skp1 19 15.3 Q9WTX6 Cullin-1
 Cul1 90 11.1 Q9WUA2 Phenylalanine--tRNA ligase beta subunit Farsb 66 7.8 Q9WUA3 ATP-dependent
 6-phosphofructokinase, platelet type Pfkp 85 9.6 Q9WUA6 RAC-gamma serine/threonine-protein kinase
 Akt3 56 4.6 Q9WUK2 Eukaryotic translation initiation factor 4H Eif4h 27 16.9 Q9WUM3 Coronin-1B
 Coro1b 54 3.7 Q9WUM4 Coronin-1C Coro1c 53 13.7 Q9WV32 Actin-related protein 2/3 complex
 subunit 1B Arpc1b 41 11 Q9WV55 Vesicle-associated membrane protein-associated Vapa 28 8.4 protein
 A Q9WV60 Glycogen synthase kinase-3 beta Gsk3b 47 19.5 Q9WVA3 Mitotic checkpoint protein BUB3
 Bub3 37 15.6 Q9WVA4 Transgelin-2 Tagln2 22 27.1 Q9Z0H8 CAP-Gly domain-containing linker
 protein 2 Clip2 116 4.8 Q9Z0N2 Eukaryotic translation initiation factor 2 subunit 3, Y- Eif2s3y 51 16.7
 linked Q9Z0P5 Twinfilin-2 Twf2 39 5.2 Q9Z130 Heterogeneous nuclear ribonucleoprotein D-like
 Hnrnpdl 34 10.6 Q9Z140 Copine-6 Cpne6 62 9.3 Q9Z172 Small ubiquitin-related modifier 3 Sumo3 12
 19.1 Q9Z1B3 1-phosphatidylinositol 4,5-bisphosphate Plcb1 138 19 phosphodiesterase beta-1 Q9Z1B7
 Mitogen-activated protein kinase 13 Mapk13 42 5.5 Q9Z1G3 V-type proton ATPase subunit C 1
 Atp6v1c1 44 7.9 Q9Z1G4 V-type proton ATPase 116 kDa subunit a isoform 1 Atp6v0a1 96 2.4 Q9Z1N5
 Spliceosome RNA helicase Ddx39b Ddx39b 49 26.4 Q9Z1S5 Neuronal-specific septin-3 Sept3 40 17.7
 Q9Z1Z2 Serine-threonine kinase receptor-associated protein Strap 38 12.3 Q9Z2H5 Band 4.1-like protein
 1 Epb41l1 98 3 Q9Z2U0 Proteasome subunit alpha type-7 Psma7 28 29 Q9Z2U1 Proteasome subunit
 alpha type-5 Psma5 26 27 Q9Z2Y8 Proline synthase co-transcribed bacterial homolog Prosc 30 12
 protein

(107) One important object of the invention is the possibility of testing candidate inhibitors in crude
 protein extracts containing native inherent TCP activity. By native inherent activity is understood the
 naturally obtained enzymatic activity contained within the biological sample and which has been obtained
 solely by described extraction method from a specific tissue, organ of biological sample. It is important to
 clarify that native activity is in a natural, unadorned or unchanged state. It has not been engineered nor
 adapted and reflects physiologically present activity in the studied biological sample such as but not
 limited to a specific tissue/organ. On the other hand, purified recombinant proteins of the different
 identified TCPase can also be tested in the dephosphorylation assay.

Example 2: Validation of the TCP Activity

(108) Before testing the different compounds of the fraction of proteins having a TCP activity (MAPs
 fraction), the TCP activity was assessed by use of a dephosphorylation assay (FIG. 4).

Material & Method

(109) An assay involving radioactively labelled tyrosine (^{35}S -Y) was used to quantitatively determine the TCPase activity. Brain tubulin was isolated and purified and radioactively labelled with ^{35}S -Y by recombinant TTL. To do so, recombinant bacterially expressed purified TTL (>90% purity) was put in contact with micrograms of pig brain purified tubulin. The reaction sample was incubated for one hour at 37° C. in presence of radioactively labelled ^{35}S -Y and ATP. Following incorporation of ^{35}S -Y to tubulin, a polymerization cycle was performed by adding GTP and incubating for 30 min at 37° C. Next the samples are centrifuged and the obtained pellet was washed twice with PEM buffer. The resulting radioactively labelled MTs were stored at -80° C. until further testing. The candidate TCPase protein was expressed and purified from bacteria using His-tagged purification strategy to at least 80% purity. After obtaining both purified recombinant TCPase candidate and radioactively labelled MTs, the proteins were put into contact and various amounts of TCPase were presented to MTs. Removal of the radioactive tyrosine by the candidate dephosphorylase was measured by quantification of radioactivity in both the soluble and insoluble fraction of the reaction using a liquid scintillation counter.

(110) To further gain insights in the potential protease that could embody the TCPase activity, the isolated MAPs from crude brain extracts were exposed to a selection of specific Cysteine, Aspartic, Metallo and Serine proteinase inhibitors were tested on the extract.

Results

(111) As expected, when MAPs fraction was added to the microtubule sample, increased dephosphorylation could be observed by western blot (Δ 1-tubulin), showing that the MAPs fraction contains dephosphorylation activity (FIG. 4). In line with previous observation, TCPase activity was found in protein extracts obtained from pig brain confirming that brain has high endogenous TCPase activity, likely in part originating from neurones.

(112) While Serine protease inhibitors did hardly reduce native TCPase activity in brain MAPs, metallo protease inhibitors, such as the EDTA and EGTA chelators, all led to significant inhibition up to 50% of the total TCPase activity contained in the brain MAPs. However, cysteine protease inhibitor treatment with compounds as E64 and Iodoacetamide (irreversible cysteine inhibitor) led to a complete inhibition of TCPase activity (FIG. 5). Of note, based on these data it seems likely that various specific proteins contain TCPase activity.

Example 3: Study of the CRMP Family

(113) As part of the identified proteins in Table I, the family of collapsing response mediator proteins (CRMPs) was found. The CRMPs family has not yet been associated with regulation of TCPase activity.

(114) In order to establish the involvement in regulation of TCP activity of these proteins, an assay was performed with a protein extract obtained from HEK293 ectopically expressing individual HA-tagged CRMP family members. All five members of the CRMP family were cloned into pRK5-HA vector and equal amounts of plasmids were transfected into HEK293 cells in a 6-wells plate. Two days after transfection cells were collected in Laemmli lysis buffer and subjected to immunoblotting analysis using a specific Δ 1-tubulin antibody. Overexpression of CRMP1 drastically increased tubulin dephosphorylation, whereas the other CRMPs showed no dephosphorylation activity (FIG. 6).

(115) To further validate the TCPase activity observed in the overexpression experiment for CRMP1, we performed immunofluorescence analysis on U2OS cells. Equal amounts of pRK5 plasmid containing all the 5 members of the CRMP family were transfected using a polyethylenimine derivative transfection reagent. Two days after transfection cells were ethanol fixed and subjected to immunofluorescence labelling of HA tag and Δ 1-tubulin (FIG. 7). All ectopically expressed CRMP proteins are labeled with HA but only CRMP1 transfected cells showed increased Δ 1-tubulin staining.

(116) An assay for evaluating knockdown of endogenous CRMP1 expression in U2OS cells by siRNA interference was also performed. U2OS cells were routinely cultured in the laboratory under standard conditions. Knockdown was obtained by transfection using INTERFERin (Polyplus) of specific RNAi sequences targeting CRMP1. Seventy two hours post-transfection, cells were collected and lysed in Laemmli buffer. Equal amounts of proteins were subjected to immunoblotting protocol. A decrease in posttranslational modification of tubulin as dephosphorylation and acetylation are shown to correlate with decrease in CRMP1 expression (FIG. 8).

(117) To further validate the loss of TCPase activity observed by knockdown of CRMP1 in U2OS cells, routinely cultured cells were plated in 6-wells plate, ethanol fixed and analysed by immunofluorescence

labelling. As anticipated, knockdown of CRMP1 resulted in reduction of $\Delta 1$ -tubulin staining (FIG. 9). (118) Interestingly, depletion of CRMP1 by transfection with specific RNAi sequences resulted in cell cycle arrest in the human U2OS cells. This was observed by western blot analysis of depleted cells. Knockdown of CRMP1 resulted in increase of p21 and p53 protein levels, a marker for cell cycle arrest (FIG. 10A). This was further validated by flow cytometry analysis. Indeed, CRMP1 depleted cells showed an increase of cells with 2n bulk DNA content, indicative for a G1 arrest (FIG. 10B-C). This is particularly interesting in the context of cancerous cells that have uncontrolled cell division.

Example 4: Inhibitor Activity of Peptidic Based Inhibitors

(119) One important object of the invention is the possibility of testing candidate inhibitor in crude protein extracts containing native inherent TCP activity. By native inherent activity is understood the naturally obtained enzymatic activity contained within the biological sample and which has been obtained solely by described extraction method from a specific tissue, organ of biological sample. According to the invention, “native activity” corresponds to natural, unadorned or unchanged state; it has not been engineered nor adapted and reflects physiologically present activity in the studied biological sample, such as but not limited to a specific tissue/organ.

(120) The natural protruding alpha tubulin tail on the surface of the MT's was used as a base for the development of peptidic inhibitors. Among the various peptides that could be tested to validate the inhibitor activity of the peptidic inhibitors of the invention, a representative set of two peptides composed of EDY and EEY was evaluated.

Material & Method

(121) Recombinant bacterially expressed purified TTL (>90% purity) was put in contact with micrograms of pig brain purified tubulin. The reaction sample was incubated for one hour at 37° C. in presence of radioactively labeled .sup.3H—Y and ATP. Following incorporation of .sup.3H—Y to tubulin, a polymerization cycle was performed by adding GTP and incubating for 30 min at 37° C. Next, the samples were centrifuged and the obtained pellet was washed twice with PEM buffer. The resulting radioactively labeled MTs were store at -80° C. until further testing. The isolated MAPs from crude brain extracts (as obtained in example 1) were contacted to the radioactively labeled MTs in absence or presence of different peptidic inhibitors or an increasing concentration of peptidic inhibitor. Release of radioactive tyrosine by native TCPase containing brain MAPs was measured by quantification of radioactivity in both the soluble and insoluble fraction of the reaction using a liquid scintillator counter.

Results

(122) This method for selecting and designing peptidic inhibitors with different properties allows differential applications of the inhibitors based on selectivity and potency criteria. We observed that the three amino acid peptide EDY already partially blocked TCP activity in the MAPs fraction (FIG. 11A). Most interestingly, the tripeptide EEY almost fully inhibited TCP activity in this setting (FIG. 11A). To further pharmacologically describe the peptidic inhibitor, a dose-response curve analysis was performed (FIG. 11B). The obtained inhibition reflected specific inhibition of TCPase by the peptidic inhibitor (FIGS. 11A and 11B).

Example 5: In Cellulo Inhibition of TCPase Activity with a Peptidic Inhibitor

(123) To further explore the in cellulo efficacy of EEY peptide to inhibit TCP activity in a relevant model, C2C12 muscle cells were cultured and differentiated.

(124) Myogenesis is a complex phenomenon and mechanistically linked to detyrosination status of the microtubules. Duchenne muscular dystrophy (DMD) is a severe type of muscular dystrophy and some of the altered biochemical processes are mimicked in the C2C12 muscle cell model.

Material & Method

(125) C2C12 cells is an immortal cell line of mouse skeletal myoblasts originally derived from satellite cells from the thigh muscle cells. C2C12 cells were routinely grown at 37° C. in a CO2 incubator. Myogenic differentiation is initiated upon reaching confluence by switching the cells to medium containing 2% horse serum. Cells were collected at start of the myogenic differentiation and every two days over a period of 6 days for molecular analysis.

(126) In addition, to assess the role of TCPase in a different model, analysis of TCPase expression in a neuronal differentiation process using neuroblastoma cell line (SH-SY5Y) was tested. SH-SY5Y cells were routinely cultured at 37° C. in a CO2 incubator prior to differentiation. Cells were resuspended in

growing media and plated at low density in culture plates. The differentiation process was followed by light microscopy and clear neural phenotype could be observed at day 8. Post-mitotic SH-SY5Y cells displayed increasing number of outgrowth and neurites. Cells were collected during the neuronal differentiation process at day 0, 2, 3, 6, 7 and 10 for gene expression analysis by quantitative PCR. qPCR probes were designed using primer3 software and CRMP1 gene expression was analyzed.

Results

(127) To further study the use of a TCPase inhibitor, C2C12 cells were treated with or without EEY peptide (FIG. 12). The protein expression of Myosin was monitored by western blotting as control for muscle differentiation. Vinculin acts as loading control. Acetylation and detyrosination of the microtubules was assessed. Whereas acetylation increases during differentiation (Ac-Tubulin) no difference in the status could be observed in the treated cells. Interestingly, detyrosination levels were increased already at day after onset of myogenic differentiation. Besides, the presence of the TCPase inhibitor clearly inhibited detyrosination (Δ -1 Tubulin), further supporting the notion that the TCPase inhibitor is cell permeable and acts on intrinsic TCPase activity.

(128) As expected a strong induction of DDC was measured during the neural differentiation process. DDC is a marker of dopaminergic neurons and validates the neural differentiation process during the experiment (FIG. 13). As anticipated, the level of CRMP1 also increased as TCPase activity also increases during the process. This is in line with the critical role of TCPase activity in the maintenance of axonal projection. The SH-SY5Y differentiation process recapitulates many molecular mechanisms known to be dysregulated in Parkinson disease and other neurodegenerative disorders.

Example 6: Study of the Detyrosination Process of Microtubules in Muscular Dystrophy

(129) Myoblast cells were obtained from a healthy control (Ctrl) and from a patient diagnosed with Duchenne muscular dystrophy (DMD), a genetic disorder characterized by progressive muscle degeneration and weakness. DMD is caused by an absence of dystrophin, a protein that helps keep muscle cells intact.

(130) The cells obtained from DMD patients have been sequenced and due to a genetic mutation, they lack dystrophin protein. After isolation and purification of the myoblast cells from the explants, the cells were cultured in a standard humidified tissue culture incubator at 37° C. in presence of 5% CO₂. The cells were amplified in Dulbecco's Modified Eagle Medium (Gibco) supplemented with 20% fetal bovine serum (FBS), 10% horse serum, purified growth factors and antibiotics (100 U/mL penicillin and 100 µg/mL streptomycin). The serum-rich growth medium supports both proliferation and differentiation of myogenic cells. The cells were plated in a 6 cm culture dish and grown to confluency prior to myogenic differentiation. Cells were collected at indicated steps by scrapping in PBS and mild centrifugation. The PBS buffer was removed and samples were snap frozen in liquid nitrogen and kept in -80° C. freezer until analysis. All the samples contained a similar amount of cells. A denaturing Laemmli buffer (containing 2% SDS, 2.5% 2-mercaptoethanol, 10% glycerol, 0.002% bromophenol blue, 0.125 M Tris HCl, pH adjusted to 6.8) was added and the samples were boiled at 95° C. to further to denature the proteins present. After cooling, the samples were loaded on a 10% polyacrylamide gel and subjected to electrophoresis for separation and transferred to a nitrocellulose membrane (GE Healthcare). The Antibodies recognizing detyrosinated tubulin (deTyr-tub), beta tubulin (E7, hybridoma) and vinculin (Sigma) were used to detect protein levels. A secondary antibody coupled to HRP (Cell Signaling) was used for detection of the protein of interest.

(131) The basal level of detyrosination was lower in DMD cells in the basal condition whereas after differentiation detyrosination was much higher (FIG. 14.). The observed increase in tubulin detyrosination in cells originating from a DMD patient could represent a new therapeutic opportunity.

Example 7: Study of the Detyrosination Process of Microtubules in Neurodegenerative Diseases

(132) a) Tauopathies belong to a class of neurodegenerative diseases associated with the pathological aggregation of the microtubule-associated protein (MAP) known as Tau protein in neurofibrillary tangles in the human brain. Tangles results from hyperphosphorylation of Tau protein, causing the protein to dissociate from microtubules and to form insoluble aggregates. Altered detyrosination will lead to exposure of negatively charged glutamate residues.

(133) The human cell line SH-SY5Y is a widely used model for studying the molecular events in the pathophysiology of Alzheimer, Parkinson disease and more generally neurodegenerative diseases. Using

SH-SY5Y, it is possible to drive differentiation in order to obtain neuronal morphology with long, extensively branched neurites that express neurospecific markers. The cells were cultured in Dulbecco's Modified Eagle Medium: Nutrient Mixture F-12 (DMEM/F-12) supplemented with 10% fetal bovine serum (FBS) in presence of antibiotics (100 U/mL penicillin and 100 µg/mL streptomycin). The cells were routinely passaged and cultured in a standard humidified tissue culture incubator at 37° C. in presence of 5% CO₂. Prior to differentiation cells were trypsinized and counted twice using an automated cell counter (Countess II; Thermo Scientific) and 0.8×10⁵ were plated in a 6-wells plate (Nunc). Next day, cells were washed twice with PBS and B-27 (Gibco) supplemented medium containing all-trans-retinoic acid (RA: Sigma R 2625) at 10 µM. Samples of SH-SY5Y cells ongoing neural differentiation were collected every day in a RIPA buffer (50 mM Tris HCl, 150 mM NaCl, 1.0% (v/v) NP-40, 0.5% (w/v) Sodium Deoxycholate, 1.0 mM EDTA, at a pH of 7.4), and quantitation of total protein performed using BCA kit (Thermo Fisher Scientific). A 20 µg protein sample of a total cell extract was run on 10% SDS-PAGE, transferred to nitrocellulose, and probed with each antibody.

(134) As observed by western blot analysis, tubulin detyrosination increases during neural differentiation. By reducing the level of detyrosination, displacement of Tau from the MT's may be hampered as such reducing intracellular aggregates. b) Recent evidences converge to the essential role of the microtubule-associated proteins known as Tau that builds up in the brain during the course of the disease but acting on microtubule modifications have so far been neglected.

(135) To further understand the role of detyrosination in the pathophysiology of Alzheimer Disease skin fibroblasts from patients carrying familial Alzheimer's Disease mutations were obtained and induced pluripotent stem cell (iPSC) generated.

(136) The cells were maintained at the neural progenitor stage and samples were collected every day in a RIPA buffer (50 mM Tris HCl, 150 mM NaCl, 1.0% (v/v) NP-40, 0.5% (w/v) Sodium Deoxycholate, 1.0 mM EDTA, at a pH of 7.4), and quantitation of total protein performed using BCA kit (Thermo Fisher Scientific). A 20 µg protein sample of a total cell extract was run on 10% SDS-PAGE, transferred to nitrocellulose, and probed with each antibody.

(137) Western blot analysis showed a striking increase of tubulin detyrosination in one of the cell-line carrying a genetic mutation (FIG. 16). Whereas current pharmaceutical efforts target the phosphorylation status of Tau protein itself, this observation opens a complete new window for therapeutic intervention. Pharmacological inhibition of detyrosinase act by directly regulating the level of microtubule detyrosination and as such may restore Tau binding to microtubules as well as endosomal-lysosomal processing efficiency, which is known to be defective during neurodegeneration.

(138) Overall inhibition of detyrosination will 1) restore axonal transport which is essential for the clearance of Tau aggregates and other aggregates, and 2) reduce the level of negatively charged amino acid at the microtubule surface (detyrosinated microtubules exposed a negatively charged glutamate) leading to improved trapping of hyperphosphorylated Tau proteins. The bulky hydrophobic aromatic residues such as tyrosine have the ability to obscure the negative charges of glutamates.

Example 8: Study of the Detyrosination Process of Microtubules in Cancers

(139) Despite the major advances in therapeutic approaches and personalized medicine, the spread of primary tumors toward distant organs and the subsequent metastatic colonization is still responsible for 90% of cancer-associated mortality. Tumors arising from epithelial tissues represent the vast majority of life-threatening cancers because of their ability to metastasize in different secondary organs. Therefore, a pressing concern in tumor biology has been the elucidation of factors and mechanisms regulating the migratory activity of these cells, tumor vascularization and colonization.

(140) Accumulating data point out that increased tumor aggressiveness is associated with misregulation of the tyrosination/detyrosination cycle of tubulin. Increased level of tubulin detyrosination has been observed during cell migration, intravasation and in colonization suggesting a key role of this modification in metastasis.

(141) By using CHL-1 cells that is a human melanoma cell line and HEK cells that have been demonstrated the ability to form colonies in soft agar and tumors of different size with varying frequencies in immunocompromised mice, we analyzed the use of a peptidic inhibitor to reduced taxol induced detyrosination. Cells were routinely cultured in a standard humidified tissue culture incubator at 37° C. in presence of 5% CO₂ and plated in a 6-wells culture dish. The cells were treated for 2 hours with

10 μ M Taxol in absence or presence of 50 μ M peptidic inhibitor.

(142) The cells were collected in a RIPA buffer (of 50 mM Tris HCl, 150 mM NaCl, 1.0% (v/v) NP-40, 0.5% (w/v) Sodium Deoxycholate, 1.0 mM EDTA, at a pH of 7.4), and quantitation of total protein performed using BCA kit (Thermo Fisher Scientific). A 20 μ g protein sample of a total cell extract was run on 10% SDS-PAGE, transferred to nitrocellulose, and probed with each antibody.

(143) Western blot analysis showed a striking decrease of taxol treated (2 hours) and consequent tubulin detyrosination in both CHL-1 and HEK cells (FIG. 17).

Claims

1. A method of treatment of a disorder involving altered microtubule detyrosination in a subject in need thereof comprising administering to the subject in need thereof an amino-acid or peptidic based inhibitor that inhibits, at least partially, a tubulin carboxypeptidase activity, wherein the amino acid or peptidic based inhibitor is selected from the group consisting of: a Tyr amino-acid (Y) modified with an epoxysuccinyl group, or a peptidic moiety consisting of an amino acid sequence selected from the group consisting of EDY, and EEY, where the tyrosine residue is chemically modified with an epoxysuccinyl group; a peptidic moiety of 2 to 19 amino acids of the most C-terminal amino acids of SEQ ID NO:12 where the C-terminal position is a tyrosine residue chemically modified with an epoxysuccinyl group; and a peptidic moiety consisting of an amino acid sequence selected from the group consisting of SEQ ID NO: 12, SEQ ID NO:13, SEQ ID NO: 14, SEQ ID NO: 15, SEQ ID NO:17, and SEQ ID NO: 18, where the C-terminal position is a tyrosine residue chemically modified with an epoxysuccinyl group.
 2. The method of claim 1, wherein the amino-acid based inhibitor is Epoxysuccinyl-Y (Eps-Y).
 3. The method of claim 1, wherein the disorder is selected from neurodegenerative diseases, cancers, muscular dystrophies, heart diseases, vascular disorders, infertility, retinal degeneration and ciliopathies.
 4. The method of claim 2, wherein the disorder is selected from neurodegenerative diseases, cancers, muscular dystrophies, heart diseases, vascular disorders, infertility, retinal degeneration and ciliopathies.
 5. The method of claim 4, wherein the neurodegenerative diseases are selected from Alzheimer's disease, Parkinson's disease, psychiatric disorders, and neural disorders, and cancers are selected from colon cancer and neuroblastoma.
 6. The method of claim 4, wherein the neurodegenerative diseases are selected from Alzheimer's disease and neural disorders and cancers are selected from colon cancer and neuroblastoma.
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